IDC® - Bearings
by INTERPRECISE Germany

# Full complement cylindrical roller bearings

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IDC®

**Full complement cylindrical roller bearings**

# General Information

Full complement cylindrical roller bearings have solid inner and outer rings. By incorporating the largest possible number of rollers, they are suitable for very heavy radial loads and are space-saving at the same time. Moreover these bearings have a high rigidity.
As a consequence of higher friction between the rollers, due to their kinematic conditions, the maximum rotational speed of full complement cylindrical roller bearings is significantly lower than that of caged cylindrical roller bearings.

# Bearing Types

As a standard, IDC® offers the full complement cylindrical roller bearings single-rowed and double-rowed. They are applicable as Floating Bearings, Support Bearings, and Locating Bearings. All bearings except Type NNF are open and can be lubricated with either oil or grease. The single row types can be lubricated through the front sides and the double row types can be lubricated through a lubrication groove or through lubrication holes. The NNF type bearings are sealed on both sides and filled with grease.

## Floating Bearings

Floating Bearings have just radial load carrying capacity.

### Type NNCL

The inner ring of this double row type has three fixed flanges, while the outer ring is flangeless. A retaining ring is inserted in the outer ring between the roller rows and keeps all bearing components together. Axial displacement of the shaft relative to the housing in both directions can be accommodated within the bearing. The retaining ring is not suitable for carrying axial loads.



## Support Bearings

Beside a high radial load carrying capacity, Support Bearings have also a small axial load carrying capacity in one direction, enabling the bearing to locate a shaft unidirectionally. In the other direction such bearings act as Floating Bearings.

### Type NCF



Bearings of this type are single rowed. They have an inner ring with two fixed flanges and an outer ring with just one fixed flange. On the flangeless side of the outer ring there is a retaining ring inserted, which keeps all bearing components together. This retaining ring serves solely for handling purposes and may not be exposed to axial load.

Due to the axial relocating ability of its outer ring this type of bearing is used e.g. to compensate length changes of shafts as a consequence of thermal expansion. The maximum axial displacement value is shown in the dimension table.

### Type NJG



In this single row design there is only the heavy series 23. This type is mainly designed for applications with extremely high radial load at low rotational speed. Bearings of this type have one outer ring with two fixed flanges and one inner ring with only one fixed flange.

In contrast to all other full complement cylindrical roller bearings, bearings of this type have a self-retaining set of rollers by default. Thus the rollers can not fall out even if the inner ring is removed. This simplifies the assembly significantly.

### Type NNCF



This type is a double row cylindrical roller bearing. The inner ring has three fixed flanges. The outer ring has only one fixed flange. On the flangeless side of the outer ring there is a retaining ring inserted which keeps all bearing components together.

These bearings are able to compensate for length changes of shafts as a consequence of thermal expansion within the range of the axial displacement ability of their outer ring.

## Locating Bearings

In addition to its high radial load carrying capacity this type of bearing is also capable of carrying axial load in both directions. This enables the bearing to guide shafts in both axial directions.

### Type NNC

The double row designed bearings of this type have an inner ring with three fixed flanges and one outer ring with two fixed flanges on the outside. A unique aspect of this is that the outer ring is axially divided in the middle. It is kept together with retaining devices. These retaining devices shall not be exposed to axial load. Therefore the outer ring has to be supported accordingly in order to be able to carry axial load.



### Type NNF

NNF bearings are always sealed on both sides by standard. The outer ring has a fixed middle flange and the inner ring has three fixed flanges. The inner ring is two-pieced and kept together by retaining rings, which shall not be exposed to axial load. Therefore the inner ring has to be supported accordingly in order to be able to carry axial load.

Since NNF bearings are used predominantly in rope sheaves and wheels the outer ring is 1mm narrower than the inner ring.

The bearings have a contact seal made from polyurethane on both sides and are retained on the inner ring shoulders to provide sufficient sealing at this position. The outer sealing lip exerts a slight pressure on the outer ring raceway. NNF bearings are filled with lithium soap grease. The operating temperature for the seals and the used grease ranges from -20°C to +80°C.

Under certain conditions, especially if operated at higher rotational speed, under higher humidity, in salt water environment, etc., the NNF bearings must be relubricated. This can be done through the inner but also through the outer ring.



Full complement
cylindrical roller bearings

# General Bearing Data

## Dimensions

The boundary dimensions of the full complement cylindrical roller bearings comply with DIN 616 and ISO 15 respectively. In case of the double row cylindrical roller bearing series 48 and 49 the dimension specifications according to DIN 5412-9 are met. However, NNF type bearings are excluded hereof.

## Operating Temperature

Standard full complement cylindrical roller bearings are suitable for operating temperatures from -30°C to +120°C. On request these bearings can be delivered heat-stabilized.
Restricted by the grease and the seal material NNF bearings are suitable for operating temperatures from -20°C to +80°C.

## Heat Treatment Process

Rings and rollers are through hardened by standard. On request the bearings can be offered with bainite or case hardened components.

## Surface

By standard rings and rollers are uncoated. On request anti-corrosion or friction reduction coatings are possible.

## Misalignment

With single row full complement cylindrical roller bearings maximum misalignment between the inner ring and the outer ring of 2 minutes of arc has no impact on the life time of the bearing.
Double row full complement cylindrical roller bearings are not able to absorb misalignment.

## Tolerances

By default full complement cylindrical roller bearings are produced to Tolerance Class PN according to DIN 620-2 and ISO 492 respectively. On request the bearings can be produced in other tolerance classes as well.

## Internal Clearance

The radial internal clearance of the full complement cylindrical roller bearings corresponds to Internal Clearance Group CN according to DIN 620-4 and ISO 5753 respectively. On request the bearings can be produced with other internal clearance as well.

| Bore d [mm] | | Radial Internal Clearance [µm] | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | CN | | C3 | | C4 | | C5 | |
| over | incl. | min | max | min | max | min | max | min | max |
| - | 24 | 20 | 45 | 35 | 60 | 50 | 75 | 65 | 90 |
| 24 | 30 | 20 | 45 | 35 | 60 | 50 | 75 | 70 | 95 |
| 30 | 40 | 25 | 50 | 45 | 70 | 60 | 85 | 80 | 105 |
| 40 | 50 | 30 | 60 | 50 | 80 | 70 | 100 | 95 | 125 |
| 50 | 65 | 40 | 70 | 60 | 90 | 80 | 110 | 110 | 140 |
| 65 | 80 | 40 | 75 | 65 | 100 | 90 | 125 | 130 | 165 |
| 80 | 100 | 50 | 85 | 75 | 110 | 105 | 140 | 155 | 190 |
| 100 | 120 | 50 | 90 | 85 | 125 | 125 | 165 | 180 | 220 |
| 120 | 140 | 60 | 105 | 100 | 145 | 145 | 190 | 200 | 245 |
| 140 | 160 | 70 | 120 | 115 | 165 | 165 | 215 | 225 | 275 |
| 160 | 180 | 75 | 125 | 120 | 170 | 170 | 220 | 250 | 300 |
| 180 | 200 | 90 | 145 | 140 | 195 | 195 | 250 | 275 | 330 |
| 200 | 225 | 105 | 165 | 160 | 220 | 220 | 280 | 305 | 365 |
| 225 | 250 | 110 | 175 | 170 | 235 | 235 | 300 | 330 | 395 |
| 250 | 280 | 125 | 195 | 190 | 260 | 260 | 330 | 370 | 440 |
| 280 | 315 | 130 | 205 | 200 | 275 | 275 | 350 | 410 | 485 |
| 315 | 355 | 145 | 225 | 225 | 305 | 305 | 385 | 455 | 535 |
| 355 | 400 | 190 | 280 | 280 | 370 | 370 | 460 | 510 | 600 |
| 400 | 450 | 210 | 310 | 310 | 410 | 410 | 510 | 565 | 665 |
| 450 | 500 | 220 | 330 | 330 | 440 | 440 | 550 | 625 | 735 |
| 500 | 560 | 240 | 360 | 360 | 480 | 480 | 600 | 690 | 810 |
| 560 | 630 | 260 | 380 | 380 | 500 | 500 | 620 | 780 | 900 |
| 630 | 710 | 285 | 425 | 425 | 565 | 565 | 705 | 865 | 1005 |
| 710 | 800 | 310 | 470 | 470 | 630 | 630 | 790 | 975 | 1135 |
| 800 | 900 | 350 | 520 | 520 | 690 | 690 | 860 | 1095 | 1265 |
| 900 | 1000 | 390 | 580 | 580 | 770 | 770 | 960 | 1215 | 1405 |
| 1000 | 1120 | 430 | 640 | 640 | 850 | 850 | 1060 | 1355 | 1565 |
| 1120 | 1250 | 470 | 710 | 710 | 950 | 950 | 1190 | 1510 | 1750 |
| 1250 | 1400 | 530 | 790 | 790 | 1050 | 1050 | 1310 | 1680 | 1940 |

## Axial Displacement

Depending on the respective series the outer ring's axial displacement relative to the inner ring in one or in both directions can be accommodated within the bearing. However, Locating Bearings are excluded. The values for the axial displacement are shown in the respective product table.

## Minimum Radial Load

With full complement cylindrical roller bearings a minimum radial load of 4% of the dynamic load rating is needed in order to ensure a failure-free operation.

## Equivalent Dynamic Bearing Load

In case of solely radial load acting on a full complement cylindrical roller bearing the dynamic load is calculated from the following equation:

$$P = F_r$$

In case full complement cylindrical roller bearings are subject to a simultaneously acting axial load the following calculation applies:

$$\frac{F_a}{F_r} \leq e \qquad P = F_r$$

$$\frac{F_a}{F_r} > e \qquad P = X * F_r + Y * F_a$$

Explanation:

e limiting value
0.15 for all double row full complement cylindrical roller bearings
0.2 for all bearings of series 18
0.3 for all other single row full complement cylindrical roller bearings

X 0.92 = radial load factor for all full complement cylindrical roller bearings

Y axial load factor
0.6 for all bearings of series 18
0.4 for all other single row full complement cylindrical roller bearings
0.53 for all double row full complement cylindrical roller bearings

In order to ensure an acceptable run of the bearing the following ratios must not be exceeded:

single row bearings: $F_a/F_r = 0.50$
double row bearings: $F_a/F_r = 0.25$

## Equivalent Static Bearing Load

In case full complement cylindrical roller bearings are exposed to static load the following calculation applies:

$$P_0 = F_r$$

## Axial Load Carrying Capacity

Full complement cylindrical roller bearings which are designed as Supporting or Locating Bearings can also accommodate axial loads. The axial load carrying capacity, however, is primarily not dependent on the material strength but rather on:

- the load carrying capacity of the sliding surfaces on the face area of the rollers and the ribs
- the lubrication of the contact areas

On basis of the following equation and parameters the permissible, permanently acting axial load can be calculated with sufficient accuracy from:

$$F_{azul} = \frac{k_1 * C_0 * 10^4}{n * (d + D)} - k_2 * F_r$$

Explanation:

| | | |
|---|---|---|
| $F_{azul}$ | highest permissible, permanently acting axial load [kN] | |
| $C_0$ | basic static load rating [kN] | |
| $F_r$ | actual radial load [kN] | |
| n | rotational speed [r/min] | |
| d | bearing bore diameter [mm] | |
| D | bearing outside diameter [mm] | |
| $k_1$ | factor dependent on the bearing type and the lubrication method: | |
| | single row bearings: | 1.00 for oil lubrication |
| | | 0.50 for grease lubrication |
| | double row bearings: | 0.35 for oil lubrication |
| | | 0.20 for grease lubrication |
| $k_2$ | factor dependent on the bearing type and the lubrication method: | |
| | single row bearings: | 0.30 for oil lubrication |
| | | 0.15 for grease lubrication |
| | double row bearings: | 0.10 for oil lubrication |
| | | 0.06 for grease lubrication |

The values for $F_{azul}$ obtained from the equation are valid for the following conditions:

- the maximum temperature difference between bearing operating temperature and environmental temperature of 60°C
- the minimum viscosity factor of 2
- the specific heat dissipation relating to the lateral area of the bearing of 0.5 mW/mm$^2$

In case grease is used as lubricant, the viscosity of the base oil has to be considered. Viscosity factors less than 2 will result in higher wear values and friction values. In order to lower these values, oils with wear protection or with respective EP additives may be used but at lower rotational speeds.

In case of axial loads acting over a longer period of time on a bearing, which is grease-lubricated, we recommend the use of grease according to DIN 51817 with a least 3% oil separation. Frequent relubrication of the bearing is recommended as well.

In case of short period acting or alternating shock acting axial loads, the following values are valid for the permissible axial load:
short period loading: $F_{akzul} = 2*F_{azul}$
alternating shock loading: $F_{aszul} = 3*F_{azul}$

However, the above values apply provided that the following limit values for axial loads, with respect to the strength of the ribs, are not exceeded:

permanent loading: $F_{amax} = 0.0023*D^{1,7}$
short period / occasional loading: $F_{amax} = 0.007*D^{1,7}$

## Radial Fastening

Normally type NNF bearings are exposed to circumferential load (e.g. with rope sheaves). For this reason the outer rings have to be located by means of interference fit in the housing.

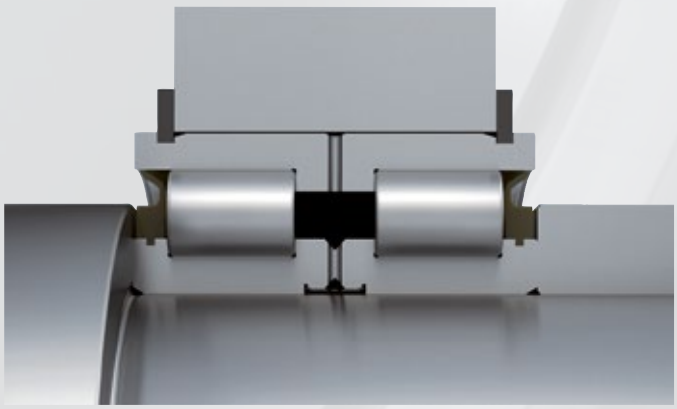
## Axial Fastening

Bearing rings have to be located with positive contact in order to avoid the axial displacement in the close-by components (shaft and housing). In this context the recommendations according to DIN 5418 shall be considered. Recesses according to DIN 509 can also be applied. In either case the minimum chamfers according to the respective product table have to be considered.

Axially loaded ribs should be supported as possible over their overall height.

Double row type NNF bearings can be fixed by simple snap rings being put into the snap ring grooves on the outer ring. By standard these snap rings are not included, however, they can be included on request (suffix 2NR).

The sealing rings of the NNF type bearings must be sufficiently supported as well. Otherwise they are likely to be pressed out when the bearing is relubricated.



## Suffixes

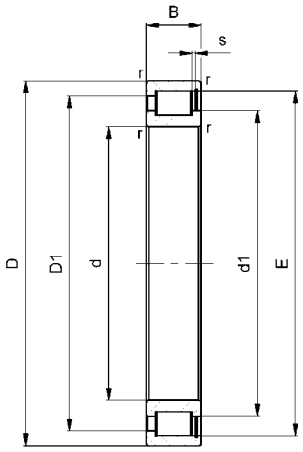
This list shows suffixes for deliverable designs which differ from the standard:

| | |
|---|---|
| 2NR | two separately included snap rings type WRE (bearing type NNF)[*1] |
| BR | burnished[*1] |
| C3 | radial internal clearance higher than CN[*1] |
| C4 | radial internal clearance higher than C3[*1] |
| C5 | radial internal clearance higher than C4[*1] |
| HA1 | inner ring and outer ring from case hardening steel[*1] |
| HB1 | inner ring and outer ring from bainite hardened[*1] |
| P6 | dimensional and running tolerance less than PN[*1] |
| P5 | dimensional and running tolerance less than P6[*1] |
| PH | phosphated[*1] |
| PP | sealing rings on both sides |
| S0 | heat stabilized up to 150°C |
| S1 | heat stabilized up to 200°C |
| S2 | heat stabilized up to 250°C |
| V | full complement |
| VH | full complement, self-retaining set of rollers |
| ZP | bearing rings with anti-corrosion coating[*1] |

[*1] on request

**Series**
**NCF 18xx.V**



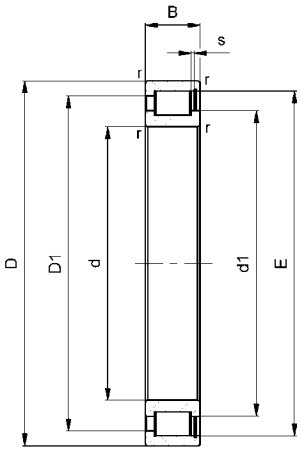



| | | measures [mm] | | | | | | | | load ratings | | fatigue limit load | limiting speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| description | weight [kg] | d | D | B | r min. | E | $d_1$ ca. | $D_1$ ca. | s[1)] | dyn. $C_r$ [kN] | stat. $C_{or}$ [kN] | $C_{ur}$ [kN] | $n_G$ [min$^{-1}$] |
| NCF1830V | 1,3 | 150 | 190 | 20 | 1,1 | 179,2 | 163,0 | 175,5 | 1,5 | 108 | 197 | 19,4 | 1900 |
| NCF1832V | 1,4 | 160 | 200 | 20 | 1,1 | 188,8 | 172,5 | 185,0 | 1,5 | 112 | 209 | 20 | 1800 |
| NCF1834V | 1,8 | 170 | 215 | 22 | 1,1 | 203,6 | 184,0 | 199,0 | 1,5 | 149 | 272 | 27,3 | 1700 |
| NCF1836V | 1,9 | 180 | 225 | 22 | 1,1 | 215,1 | 195,5 | 210,5 | 1,5 | 154 | 290 | 28,3 | 1600 |
| NCF1838V | 2,4 | 190 | 240 | 24 | 1,5 | 228,8 | 208,0 | 224,0 | 1,8 | 176 | 330 | 31,5 | 1500 |
| NCF1840V | 2,6 | 200 | 250 | 24 | 1,5 | 237,1 | 216,0 | 232,0 | 1,8 | 179 | 343 | 33 | 1400 |
| NCF1844V | 2,8 | 220 | 270 | 24 | 1,5 | 257,9 | 237,0 | 253,0 | 1,8 | 188 | 377 | 35,5 | 1200 |
| NCF1848V | 4,4 | 240 | 300 | 28 | 2 | 286,7 | 261,0 | 280,5 | 1,8 | 264 | 520 | 49,5 | 1100 |
| NCF1852V | 4,7 | 260 | 320 | 28 | 2 | 307,1 | 281,0 | 301,0 | 1,8 | 275 | 561 | 52,5 | 1000 |
| NCF1856V | 7,1 | 280 | 350 | 33 | 2 | 334,0 | 305,0 | 327,0 | 2,5 | 335 | 682 | 64 | 950 |
| NCF1860V | 10 | 300 | 380 | 38 | 2,1 | 362,9 | 329,0 | 355,0 | 3 | 428 | 858 | 79 | 850 |
| NCF1864V | 10,5 | 320 | 400 | 38 | 2,1 | 383,0 | 349,0 | 375,0 | 3 | 442 | 912 | 82 | 800 |
| NCF1868V | 11 | 340 | 420 | 38 | 2,1 | 403,1 | 369,0 | 395,0 | 3 | 455 | 966 | 86 | 750 |
| NCF1872V | 11,5 | 360 | 440 | 38 | 2,1 | 423,3 | 389,0 | 415,0 | 3,5 | 468 | 1020 | 92 | 700 |
| NCF1876V | 19,5 | 380 | 480 | 46 | 2,1 | 457,7 | 415,5 | 447,0 | 3,5 | 651 | 1360 | 122 | 670 |
| NCF1880V | 20,5 | 400 | 500 | 46 | 2,1 | 474,2 | 432,0 | 464,0 | 3,5 | 664 | 1420 | 127 | 630 |
| NCF1884V | 21 | 420 | 520 | 46 | 2,1 | 499,1 | 457,0 | 489,0 | 3,5 | 684 | 1500 | 131 | 600 |
| NCF1888V | 22 | 440 | 540 | 46 | 2,1 | 515,7 | 473,5 | 505,0 | 3,5 | 697 | 1560 | 134 | 560 |
| NCF1892V | 34 | 460 | 580 | 56 | 3 | 553,0 | 501,0 | 540,0 | 5 | 945 | 2020 | 171 | 530 |
| NCF1896V | 35,5 | 480 | 600 | 56 | 3 | 573,5 | 522,0 | 561,0 | 5 | 965 | 2100 | 180 | 500 |
| NCF18/500V | 36,5 | 500 | 620 | 56 | 3 | 594,0 | 542,0 | 581,0 | 5 | 985 | 2190 | 179 | 480 |
| NCF18/530V | 38,5 | 530 | 650 | 56 | 3 | 624,5 | 573,0 | 612,0 | 5 | 1010 | 2320 | 184 | 450 |
| NCF18/560V | 40,5 | 560 | 680 | 56 | 3 | 655,0 | 603,0 | 642,0 | 5 | 1040 | 2440 | 190 | 430 |
| NCF18/600V | 51,5 | 600 | 730 | 60 | 3 | 696,0 | 644,0 | 683,0 | 7 | 1080 | 2610 | 200 | 400 |
| NCF18/630V | 72,5 | 630 | 780 | 69 | 4 | 737,0 | 679,0 | 723,0 | 8 | 1290 | 3080 | 235 | 360 |
| NCF18/670V | 76,5 | 670 | 820 | 69 | 4 | 783,0 | 725,0 | 768,0 | 8 | 1340 | 3290 | 250 | 340 |
| NCF18/710V | 92,5 | 710 | 870 | 74 | 4 | 831,5 | 767,0 | 816,0 | 8 | 1590 | 3890 | 290 | 320 |
| NCF18/750V | 110 | 750 | 920 | 78 | 5 | 880,0 | 811,0 | 863,0 | 8 | 1790 | 4380 | 320 | 300 |
| NCF18/800V | 130 | 800 | 980 | 82 | 5 | 935,5 | 863,0 | 917,0 | 9 | 1970 | 4890 | 350 | 280 |
| NCF18/850V | 135 | 850 | 1030 | 82 | 5 | 985,5 | 911,0 | 967,0 | 9 | 2090 | 5290 | 375 | 260 |
| NCF18/900V | 160 | 900 | 1090 | 85 | 5 | 1044,0 | 966,0 | 1024,0 | 9 | 2280 | 5880 | 405 | 240 |
| NCF18/950V | 185 | 950 | 1150 | 90 | 5 | 1103,0 | 1022,0 | 1083,0 | 10 | 2490 | 6500 | 450 | 220 |
| NCF18/1000V | 230 | 1000 | 1220 | 100 | 6 | 1164,0 | 1074,0 | 1142,0 | 12 | 2980 | 7690 | 465 | 200 |

[1)] axial displacement facility from central position

## Series
## NCF 22xx.V



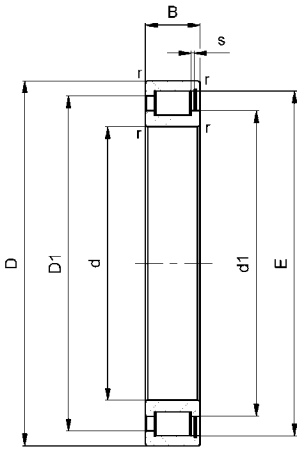



| | | measures [mm] | | | | | | | | load ratings | | fatigue limit load | limiting speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| description | weight [kg] | d | D | B | r min. | E | $d_1$ ca. | $D_1$ ca. | s[1] | dyn. $C_r$ [kN] | stat. $C_{or}$ [kN] | $C_{ur}$ [kN] | $n_G$ [min$^{-1}$] |
| NCF2204V | 0,16 | 20 | 47 | 18 | 1 | 41,6 | 28,7 | 38,3 | 1 | 40 | 39 | 5,4 | 9700 |
| NCF2205V | 0,18 | 25 | 52 | 18 | 1 | 46,6 | 33,8 | 43,4 | 1 | 45 | 47 | 6,6 | 8400 |
| NCF2206V | 0,30 | 30 | 62 | 20 | 1 | 55,2 | 40,8 | 51,6 | 1 | 61 | 66 | 8,9 | 7000 |
| NCF2207V | 0,44 | 35 | 72 | 23 | 1,1 | 64,0 | 46,4 | 59,6 | 1 | 76 | 81 | 10,5 | 6100 |
| NCF2208V | 0,55 | 40 | 80 | 23 | 1,1 | 71,0 | 53,4 | 66,6 | 1 | 84 | 94 | 13 | 5400 |
| NCF2209V | 0,59 | 45 | 85 | 23 | 1,1 | 74,5 | 56,9 | 70,1 | 1 | 87 | 101 | 13,5 | 5000 |
| NCF2210V | 0,64 | 50 | 90 | 23 | 1,1 | 81,5 | 63,9 | 77,1 | 1 | 94 | 114 | 15 | 4650 |
| NCF2211V | 0,87 | 55 | 100 | 25 | 1,5 | 88,9 | 69,7 | 84,1 | 1 | 113 | 141 | 20,5 | 4200 |
| NCF2212V | 1,2 | 60 | 110 | 28 | 1,5 | 99,3 | 76,8 | 93,6 | 1,5 | 148 | 184 | 27 | 3800 |
| NCF2213V | 1,6 | 65 | 120 | 31 | 1,5 | 106,3 | 82,3 | 100,3 | 1,5 | 172 | 218 | 32 | 3500 |
| NCF2214V | 1,7 | 70 | 125 | 31 | 1,5 | 111,1 | 87,1 | 105,1 | 1,5 | 179 | 232 | 31 | 3300 |
| NCF2215V | 1,8 | 75 | 130 | 31 | 1,5 | 115,9 | 91,8 | 109,8 | 1,5 | 185 | 246 | 32,5 | 3150 |
| NCF2216V | 2,1 | 80 | 140 | 33 | 2 | 125,9 | 98,7 | 119,1 | 1,5 | 218 | 286 | 37 | 2950 |
| NCF2217V | 2,7 | 85 | 150 | 36 | 2 | 133,3 | 104,5 | 126,5 | 1,5 | 247 | 329 | 43 | 2750 |
| NCF2218V | 3,5 | 90 | 160 | 40 | 2 | 140,7 | 110,3 | 133,5 | 2,5 | 279 | 376 | 48 | 2600 |
| NCF2219V | 4,2 | 95 | 170 | 43 | 2,1 | 155,5 | 121,5 | 147,5 | 2,5 | 327 | 443 | 55,5 | 2450 |
| NCF2220V | 5,1 | 100 | 180 | 46 | 2,1 | 162,9 | 127,5 | 154,5 | 2,5 | 371 | 513 | 63,5 | 2310 |
| NCF2222V | 7,2 | 110 | 200 | 53 | 2,1 | 177,7 | 139,0 | 168,5 | 4 | 437 | 613 | 72 | 2090 |
| NCF2224V | 9 | 120 | 215 | 58 | 2,1 | 192,5 | 150,5 | 182,5 | 4 | 519 | 742 | 88,5 | 1930 |
| NCF2226V | 11,2 | 130 | 230 | 64 | 3 | 207,4 | 162,5 | 196,5 | 5 | 603 | 874 | 101 | 1800 |
| NCF2228V | 14,4 | 140 | 250 | 68 | 3 | 222,2 | 174,0 | 210,5 | 5 | 697 | 1020 | 119 | 1660 |
| NCF2230V | 18,4 | 150 | 270 | 73 | 3 | 237,0 | 185,5 | 224,5 | 6 | 797 | 1190 | 136 | 1540 |
| NCF2232V | 23 | 160 | 290 | 80 | 3 | 266,6 | 209,0 | 252,5 | 6 | 988 | 1500 | 165 | 1440 |
| NCF2234V | 28,7 | 170 | 310 | 86 | 4 | 281,4 | 220,5 | 266,5 | 7 | 1100 | 1680 | 183 | 1350 |
| NCF2236V | 29,8 | 180 | 320 | 86 | 4 | 293,5 | 232,5 | 278,5 | 7 | 1140 | 1780 | 190 | 1300 |
| NCF2238V | 35,7 | 190 | 340 | 92 | 4 | 308,9 | 244,5 | 293,0 | 7 | 1250 | 1970 | 206 | 1220 |
| NCF2240V | 43,1 | 200 | 360 | 98 | 4 | 324,4 | 257,0 | 307,5 | 7 | 1400 | 2230 | 227 | 1160 |
| NCF2244V | 58 | 220 | 400 | 108 | 4 | 355,5 | 278,5 | 336,5 | 8 | 1690 | 2680 | 235 | 1100 |

[1] axial displacement facility from central position

## Series
## NCF 29xx.V



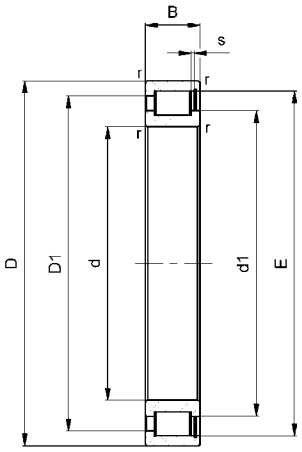



| | | measures [mm] | | | | | | | | load ratings | | fatigue limit load | limiting speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| description | weight [kg] | d | D | B | r min. | E | $d_1$ ca. | $D_1$ ca. | $s^{1)}$ | dyn. $C_r$ [kN] | stat. $C_{or}$ [kN] | $C_{ur}$ [kN] | $n_G$ [$min^{-1}$] |
| NCF2904V | 0,05 | 20 | 37 | 11 | 0,3 | 32,2 | 26,2 | 30,2 | 0,5 | 16 | 17 | 2 | 11000 |
| NCF2905V | 0,06 | 25 | 42 | 11 | 0,3 | 37,3 | 31,3 | 35,3 | 0,5 | 17 | 21 | 2,3 | 9600 |
| NCF2906V | 0,07 | 30 | 47 | 11 | 0,3 | 42,4 | 36,4 | 40,4 | 0,5 | 19 | 25 | 2,7 | 8300 |
| NCF2907V | 0,12 | 35 | 55 | 13 | 0,6 | 49,8 | 42,3 | 47,3 | 0,5 | 30 | 39 | 4,6 | 7100 |
| NCF2908V | 0,15 | 40 | 62 | 14 | 0,6 | 56,5 | 48,3 | 53,8 | 0,5 | 34 | 45 | 5,6 | 6300 |
| NCF2909V | 0,18 | 45 | 68 | 14 | 0,6 | 61,8 | 53,5 | 59,0 | 0,5 | 36 | 50 | 6 | 5600 |
| NCF2910V | 0,19 | 50 | 72 | 14 | 0,6 | 67,0 | 58,8 | 64,3 | 0,5 | 38 | 56 | 6,7 | 5200 |
| NCF2911V | 0,27 | 55 | 80 | 16 | 1 | 73,1 | 64,1 | 70,1 | 0,5 | 50 | 77 | 8,5 | 4800 |
| NCF2912V | 0,29 | 60 | 85 | 16 | 1 | 79,2 | 69,2 | 76,2 | 1 | 55 | 84 | 10,3 | 4500 |
| NCF2913V | 0,31 | 65 | 90 | 16 | 1 | 83,3 | 73,3 | 80,3 | 1 | 57 | 89 | 11,2 | 4000 |
| NCF2914V | 0,49 | 70 | 100 | 19 | 1 | 92,4 | 79,6 | 89,1 | 1 | 77 | 118 | 15,1 | 3800 |
| NCF2915V | 0,52 | 75 | 105 | 19 | 1 | 97,5 | 84,7 | 94,2 | 1 | 80 | 126 | 16,1 | 3600 |
| NCF2916V | 0,55 | 80 | 110 | 19 | 1 | 102,6 | 89,8 | 99,3 | 1 | 83 | 134 | 17,2 | 3400 |
| NCF2917V | 0,81 | 85 | 120 | 22 | 1,1 | 112,5 | 98,2 | 108,9 | 1 | 104 | 169 | 21 | 3200 |
| NCF2918V | 0,84 | 90 | 125 | 22 | 1,1 | 115,4 | 101,0 | 111,8 | 1 | 105 | 174 | 21,5 | 3000 |
| NCF2919V | 0,86 | 95 | 130 | 22 | 1,1 | 121,1 | 106,8 | 117,5 | 1 | 108 | 185 | 22 | 2800 |
| NCF2920V | 1,14 | 100 | 140 | 24 | 1,1 | 128,2 | 112,2 | 124,5 | 1,5 | 125 | 209 | 25 | 2600 |
| NCF2922V | 1,23 | 110 | 150 | 24 | 1,1 | 141,0 | 125,0 | 137,0 | 1,5 | 132 | 233 | 27,5 | 2400 |
| NCF2924V | 1,73 | 120 | 165 | 27 | 1,1 | 153,8 | 134,5 | 149,0 | 1,5 | 177 | 304 | 37,5 | 2200 |
| NCF2926V | 2,33 | 130 | 180 | 30 | 1,5 | 166,7 | 145,5 | 161,5 | 2 | 208 | 364 | 44 | 2000 |
| NCF2928V | 2,42 | 140 | 190 | 30 | 1,5 | 179,5 | 157,0 | 174,0 | 2 | 225 | 392 | 46,5 | 1900 |
| NCF2930V | 3,77 | 150 | 210 | 36 | 2 | 194,9 | 169,0 | 188,5 | 2,5 | 293 | 512 | 59 | 1700 |
| NCF2932V | 4 | 160 | 220 | 36 | 2 | 205,1 | 179,0 | 199,0 | 2,5 | 302 | 544 | 62 | 1600 |
| NCF2934V | 4,3 | 170 | 230 | 36 | 2 | 215,3 | 189,5 | 209,0 | 2,5 | 311 | 576 | 63,5 | 1500 |
| NCF2936V | 6,2 | 180 | 250 | 42 | 2 | 231,4 | 201,0 | 224,0 | 2,5 | 397 | 709 | 81,5 | 1400 |
| NCF2938V | 6,5 | 190 | 260 | 42 | 2 | 243,6 | 213,0 | 236,0 | 2,5 | 410 | 753 | 82,5 | 1400 |
| NCF2940V | 9,1 | 200 | 280 | 48 | 2,1 | 262,5 | 228,5 | 254,0 | 3 | 498 | 924 | 101 | 1300 |
| NCF2944V | 9,9 | 220 | 300 | 48 | 2,1 | 282,6 | 248,5 | 274,5 | 3 | 522 | 1010 | 108 | 1200 |
| NCF2948V | 10,6 | 240 | 320 | 48 | 2,1 | 302,7 | 268,5 | 294,5 | 3 | 544 | 1090 | 109 | 1100 |
| NCF2952V | 18,5 | 260 | 360 | 60 | 2,1 | 333,2 | 291,5 | 323,0 | 3,5 | 764 | 1490 | 150 | 950 |
| NCF2956V | 19,7 | 280 | 380 | 60 | 2,1 | 358,9 | 314,0 | 348,0 | 3,5 | 887 | 1750 | 172 | 900 |
| NCF2960V | 31,2 | 300 | 420 | 72 | 3 | 389,8 | 338,5 | 377,0 | 5 | 1130 | 2240 | 218 | 800 |
| NCF2964V | 32,9 | 320 | 440 | 72 | 3 | 410,2 | 358,5 | 397,5 | 5 | 1170 | 2380 | 228 | 750 |
| NCF2968V | 35 | 340 | 460 | 72 | 3 | 430,6 | 379,0 | 418,0 | 5 | 1210 | 2520 | 238 | 700 |
| NCF2972V | 36,5 | 360 | 480 | 72 | 3 | 451,0 | 399,5 | 438,0 | 5 | 1240 | 2660 | 250 | 670 |
| NCF2976V | 52,5 | 380 | 520 | 82 | 4 | 484,4 | 426,5 | 470,0 | 5 | 1490 | 3160 | 290 | 630 |
| NCF2980V | 54,5 | 400 | 540 | 82 | 4 | 507,5 | 449,5 | 493,0 | 5 | 1530 | 3330 | 300 | 600 |
| NCF2984V | 57 | 420 | 560 | 82 | 4 | 530,5 | 472,5 | 516,0 | 5 | 1570 | 3510 | 310 | 560 |
| NCF2988V | 80,5 | 440 | 600 | 95 | 4 | 565,0 | 497,5 | 549,0 | 6 | 2040 | 4480 | 390 | 530 |
| NCF2992V | 83,5 | 460 | 620 | 95 | 4 | 578,5 | 511,0 | 562,0 | 6 | 2070 | 4600 | 400 | 500 |
| NCF2996V | 98 | 480 | 650 | 100 | 5 | 605,5 | 533,0 | 588,0 | 7 | 2310 | 5110 | 420 | 480 |
| NCF29/500V | 100 | 500 | 670 | 100 | 5 | 634,0 | 561,0 | 616,0 | 7 | 2380 | 5400 | 440 | 450 |
| NCF29/530V | 120 | 530 | 710 | 106 | 5 | 673,0 | 592,0 | 653,0 | 7 | 2730 | 6080 | 490 | 430 |
| NCF29/560V | 140 | 560 | 750 | 112 | 5 | 709,5 | 622,0 | 688,0 | 7 | 3070 | 6790 | 500 | 400 |
| NCF29/600V | 170 | 600 | 800 | 118 | 5 | 753,5 | 663,0 | 731,0 | 7 | 3390 | 7690 | 550 | 380 |
| NCF29/630V | 205 | 630 | 850 | 128 | 6 | 807,5 | 711,0 | 783,0 | 8 | 3790 | 8650 | 620 | 340 |
| NCF29/670V | 245 | 670 | 900 | 136 | 6 | 854,0 | 754,0 | 829,0 | 10 | 4150 | 9690 | 670 | 320 |

[1)] axial displacement facility from central position

**Series**
**NCF 30xx.V**



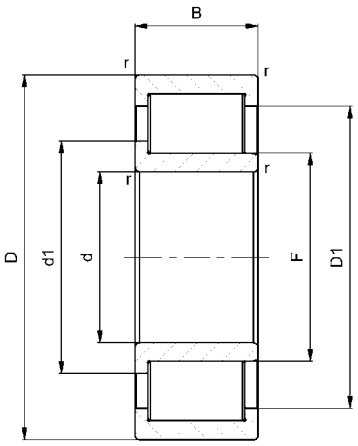



| | | measures [mm] | | | | | | | | load ratings | | fatigue limit load | limiting speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| description | weight [kg] | d | D | B | r min. | E | $d_1$ ca. | $D_1$ ca. | s[1] | dyn. $C_r$ [kN] | stat. $C_{or}$ [kN] | $C_{ur}$ [kN] | $n_G$ [min$^{-1}$] |
| NCF3004V | 0,11 | 20 | 42 | 16 | 0,6 | 37,9 | 27,5 | 35,2 | 1,5 | 28 | 28 | 3,6 | 10000 |
| NCF3005V | 0,12 | 25 | 47 | 16 | 0,6 | 42,0 | 31,6 | 39,3 | 1,5 | 31 | 33 | 4,3 | 9000 |
| NCF3006V | 0,2 | 30 | 55 | 19 | 1 | 49,7 | 38,5 | 46,8 | 2 | 40 | 45 | 5,8 | 7500 |
| NCF3007V | 0,26 | 35 | 62 | 20 | 1 | 55,6 | 43,6 | 52,5 | 2 | 49 | 57 | 7,4 | 6700 |
| NCF3008V | 0,31 | 40 | 68 | 21 | 1 | 61,8 | 49,0 | 58,5 | 2 | 58 | 70 | 9 | 6000 |
| NCF3009V | 0,4 | 45 | 75 | 23 | 1 | 68,4 | 54,8 | 65,0 | 2 | 72 | 92 | 11,8 | 5300 |
| NCF3010V | 0,43 | 50 | 80 | 23 | 1 | 73,8 | 60,2 | 70,4 | 2 | 76 | 102 | 12,4 | 5000 |
| NCF3011V | 0,64 | 55 | 90 | 26 | 1,1 | 83,6 | 67,6 | 79,6 | 2 | 98 | 131 | 17,3 | 4300 |
| NCF3012V | 0,69 | 60 | 95 | 26 | 1,1 | 86,8 | 70,8 | 82,8 | 2 | 101 | 138 | 18,4 | 4000 |
| NCF3013V | 0,73 | 65 | 100 | 26 | 1,1 | 93,2 | 77,2 | 89,1 | 2 | 107 | 151 | 20,1 | 3800 |
| NCF3014V | 1,02 | 70 | 110 | 30 | 1,1 | 102,0 | 82,5 | 97,1 | 3 | 133 | 179 | 24,5 | 3600 |
| NCF3015V | 1,06 | 75 | 115 | 30 | 1,1 | 105,9 | 86,4 | 101,0 | 3 | 137 | 188 | 26 | 3200 |
| NCF3016V | 1,43 | 80 | 125 | 34 | 1,1 | 117,1 | 94,6 | 111,4 | 4 | 168 | 230 | 30 | 3000 |
| NCF3017V | 1,51 | 85 | 130 | 34 | 1,1 | 121,5 | 99,1 | 115,9 | 4 | 173 | 242 | 31 | 3000 |
| NCF3018V | 1,97 | 90 | 140 | 37 | 1,5 | 130,2 | 106,2 | 124,5 | 4 | 202 | 287 | 36,5 | 2800 |
| NCF3020V | 2,15 | 100 | 150 | 37 | 1,5 | 139,7 | 115,7 | 134,0 | 4 | 212 | 314 | 38,5 | 2600 |
| NCF3022V | 3,5 | 110 | 170 | 45 | 2 | 156,2 | 127,0 | 149,0 | 5,5 | 280 | 407 | 49,5 | 2200 |
| NCF3024V | 3,8 | 120 | 180 | 46 | 2 | 167,7 | 138,5 | 160,5 | 5,5 | 294 | 446 | 53,5 | 2000 |
| NCF3026V | 5,8 | 130 | 200 | 52 | 2 | 184,0 | 148,5 | 175,5 | 5,5 | 418 | 624 | 74 | 1900 |
| NCF3028V | 6,1 | 140 | 210 | 53 | 2 | 197,9 | 162,5 | 189,5 | 5,5 | 441 | 686 | 80 | 1800 |
| NCF3030V | 7,5 | 150 | 225 | 56 | 2,1 | 206,9 | 170,0 | 198,0 | 7 | 463 | 717 | 83 | 1600 |
| NCF3032V | 9,1 | 160 | 240 | 60 | 2,1 | 224,9 | 184,5 | 215,0 | 7 | 530 | 628 | 94 | 1500 |
| NCF3034V | 12,5 | 170 | 260 | 67 | 2,1 | 243,0 | 198,0 | 232,0 | 7 | 681 | 1080 | 122 | 1400 |
| NCF3036V | 16,5 | 180 | 280 | 74 | 2,1 | 260,3 | 212,0 | 248,5 | 7 | 788 | 1270 | 139 | 1300 |
| NCF3038V | 17 | 190 | 290 | 75 | 2,1 | 269,9 | 221,5 | 258,0 | 9 | 809 | 1330 | 146 | 1300 |
| NCF3040V | 22,5 | 200 | 310 | 82 | 2,1 | 287,9 | 236,5 | 275,0 | 9 | 925 | 1540 | 167 | 1200 |
| NCF3044V | 29,5 | 220 | 340 | 90 | 3 | 312,4 | 254,5 | 298,0 | 9 | 1110 | 1840 | 196 | 1100 |
| NCF3048V | 32 | 240 | 360 | 92 | 3 | 335,3 | 277,5 | 321,0 | 11 | 1170 | 2010 | 210 | 1000 |
| NCF3052V | 46,5 | 260 | 400 | 104 | 4 | 376,2 | 304,0 | 358,5 | 11 | 1590 | 2640 | 265 | 900 |
| NCF3056V | 50 | 280 | 420 | 106 | 4 | 390,5 | 318,0 | 372,5 | 11 | 1630 | 2770 | 275 | 850 |
| NCF3060V | 69 | 300 | 460 | 118 | 4 | 431,8 | 354,5 | 412,5 | 14 | 2000 | 3560 | 315 | 750 |
| NCF3064V | 74,5 | 320 | 480 | 121 | 4 | 447,1 | 370,2 | 428,0 | 14 | 2050 | 3720 | 325 | 700 |
| NCF3068V | 100 | 340 | 520 | 133 | 5 | 481,6 | 401,5 | 461,5 | 14 | 2410 | 4570 | 370 | 670 |
| NCF3072V | 105 | 360 | 540 | 134 | 5 | 503,0 | 416,0 | 481,5 | 14 | 2550 | 4720 | 395 | 630 |
| NCF3076V | 110 | 380 | 560 | 135 | 5 | 520,0 | 433,5 | 498,5 | 14 | 2620 | 4930 | 410 | 600 |
| NCF3080V | 145 | 400 | 600 | 148 | 5 | 559,0 | 462,5 | 535,0 | 14 | 2980 | 5520 | 465 | 560 |
| NCF3084V | 150 | 420 | 620 | 150 | 5 | 578,0 | 481,5 | 554,0 | 15 | 3050 | 5760 | 460 | 530 |
| NCF3088V | 175 | 440 | 650 | 157 | 6 | 616,5 | 513,0 | 591,0 | 16 | 3400 | 6480 | 495 | 500 |
| NCF3092V | 195 | 460 | 680 | 163 | 6 | 633,5 | 524,0 | 606,0 | 16 | 3680 | 6910 | 540 | 480 |
| NCF3096V | 205 | 480 | 700 | 165 | 6 | 655,0 | 546,0 | 628,0 | 16 | 3770 | 7210 | 550 | 450 |
| NCF30/500V | 215 | 500 | 720 | 167 | 6 | 676,5 | 567,0 | 650,0 | 16 | 3850 | 7510 | 580 | 450 |

[1] axial displacement facility from central position

## Series
## NJG 23xx.VH



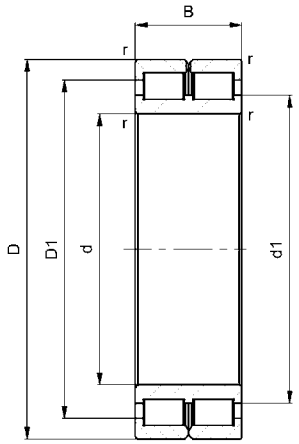



| | | measures [mm] | | | | | | | | load ratings | | fatigue limit load | limiting speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| description | weight [kg] | d | D | B | r min. | F | $d_1$ ca. | $D_1$ ca. | s[1)] | dyn. $C_r$ [kN] | stat. $C_{or}$ [kN] | $C_{ur}$ [kN] | $n_G$ [$min^{-1}$] |
| NJG2305VH | 0,38 | 25 | 62 | 24 | 1,1 | 31,7 | 36,5 | 49,0 | 1,7 | 66 | 65 | 8,4 | 5600 |
| NJG2306VH | 0,56 | 30 | 72 | 27 | 1,1 | 38,3 | 43,5 | 57,0 | 1,8 | 85 | 88 | 11,7 | 4800 |
| NJG2307VH | 0,75 | 35 | 80 | 31 | 1,5 | 44,7 | 51,0 | 66,5 | 2 | 119 | 127 | 16,8 | 4300 |
| NJG2308VH | 1 | 40 | 90 | 33 | 1,5 | 51,1 | 58,0 | 76,0 | 2,4 | 146 | 158 | 21,6 | 3600 |
| NJG2309VH | 1,4 | 45 | 100 | 36 | 1,5 | 56,1 | 63,0 | 81,0 | 2,4 | 172 | 196 | 27 | 3400 |
| NJG2310VH | 1,8 | 50 | 110 | 40 | 2 | 60,7 | 69,0 | 90,5 | 2,6 | 200 | 220 | 31 | 3000 |
| NJG2311VH | 2,3 | 55 | 120 | 43 | 2 | 67,0 | 76,0 | 100,0 | 2,6 | 234 | 260 | 36 | 2800 |
| NJG2312VH | 2,9 | 60 | 130 | 46 | 2,1 | 73,6 | 83,0 | 106,5 | 3 | 263 | 306 | 46 | 2600 |
| NJG2313VH | 3,6 | 65 | 140 | 48 | 2,1 | 80,6 | 90,5 | 117,0 | 3 | 304 | 357 | 50,5 | 2400 |
| NJG2314VH | 4,4 | 70 | 150 | 51 | 2,1 | 84,1 | 94,5 | 122,0 | 3 | 334 | 396 | 54,5 | 2200 |
| NJG2315VH | 5,3 | 75 | 160 | 55 | 2,1 | 91,1 | 102,5 | 132,0 | 3 | 396 | 479 | 65,5 | 2000 |
| NJG2316VH | 6,4 | 80 | 170 | 58 | 2,1 | 98,2 | 110,0 | 142,5 | 4 | 460 | 564 | 76,5 | 1900 |
| NJG2317VH | 7,4 | 85 | 180 | 60 | 3 | 106,9 | 119,0 | 151,5 | 4 | 487 | 615 | 82 | 1800 |
| NJG2318VH | 8,8 | 90 | 190 | 64 | 3 | 108,7 | 122,0 | 158,0 | 4 | 551 | 684 | 92 | 1700 |
| NJG2319VH | 10,2 | 95 | 200 | 67 | 3 | 112,2 | 125,5 | 163,0 | 4 | 608 | 767 | 112 | 1600 |
| NJG2320VH | 13 | 100 | 215 | 73 | 3 | 119,2 | 133,5 | 173,0 | 4 | 679 | 864 | 113 | 1500 |
| NJG2322VH | 17,5 | 110 | 240 | 80 | 3 | 133,2 | 149,5 | 193,5 | 5 | 839 | 1080 | 128 | 1300 |
| NJG2324VH | 22,5 | 120 | 260 | 86 | 3 | 147,3 | 164,5 | 214,5 | 5 | 970 | 1260 | 155 | 1200 |
| NJG2326VH | 28 | 130 | 280 | 93 | 4 | 157,8 | 176,5 | 229,5 | 6 | 1100 | 1440 | 159 | 1200 |
| NJG2328VH | 35,5 | 140 | 300 | 102 | 4 | 168,3 | 188,0 | 245,0 | 6,5 | 1250 | 1660 | 178 | 1100 |
| NJG2330VH | 42,5 | 150 | 320 | 108 | 4 | 182,4 | 204,0 | 265,0 | 6,5 | 1510 | 2040 | 203 | 1000 |
| NJG2332VH | 49 | 160 | 340 | 114 | 4 | 196,4 | 219,5 | 285,5 | 7 | 1700 | 2340 | 227 | 950 |
| NJG2334VH | 59,5 | 170 | 360 | 120 | 4 | 203,4 | 227,5 | 296,0 | 7 | 1790 | 2460 | 240 | 900 |
| NJG2336VH | 69,5 | 180 | 380 | 126 | 4 | 217,5 | 243,0 | 316,5 | 8 | 1980 | 2720 | 265 | 800 |
| NJG2338VH | 80 | 190 | 400 | 132 | 5 | 224,5 | 250,5 | 326,5 | 8 | 2130 | 2960 | 275 | 800 |
| NJG2340VH | 92 | 200 | 420 | 138 | 5 | 238,5 | 266,5 | 347,0 | 9 | 2360 | 3290 | 295 | 750 |
| NJG2344VH | 111 | 220 | 460 | 145 | 5 | 267,5 | 296,5 | 379,0 | 10 | 2680 | 3880 | 335 | 670 |
| NJG2348VH | 147 | 240 | 500 | 155 | 5 | 290,4 | 321,5 | 411,5 | 10 | 3030 | 4410 | 370 | 630 |
| NJG2352VH | 177 | 260 | 540 | 165 | 6 | 313,3 | 347,0 | 444,0 | 11 | 3380 | 4930 | 410 | 530 |

[1)] axial displacement facility from central position

## Series
## NNC 48xx.V

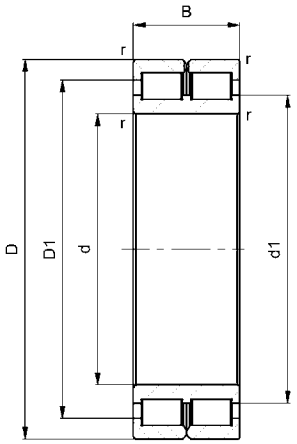





| | | measures [mm] | | | | | | load ratings | | fatigue limit load | limiting speed |
|---|---|---|---|---|---|---|---|---|---|---|---|
| description | weight [kg] | d | D | B | r min. | $d_1$ ca. | $D_1$ ca. | dyn. $C_r$ [kN] | stat. $C_{or}$ [kN] | $C_{ur}$ [kN] | $n_G$ [$min^{-1}$] |
| NNC4830V | 2,9 | 150 | 190 | 40 | 1,1 | 164,0 | 175,0 | 234 | 575 | 58 | 1910 |
| NNC4832V | 3,1 | 160 | 200 | 40 | 1,1 | 175,5 | 186,5 | 243 | 616 | 61 | 1800 |
| NNC4834V | 4,1 | 170 | 215 | 45 | 1,1 | 185,5 | 197,5 | 265 | 651 | 64 | 1680 |
| NNC4836V | 4,3 | 180 | 225 | 45 | 1,1 | 195,0 | 207,5 | 272 | 685 | 67 | 1600 |
| NNC4838V | 5,7 | 190 | 240 | 50 | 1,5 | 207,5 | 221,0 | 314 | 784 | 77 | 1510 |
| NNC4840V | 5,9 | 200 | 250 | 50 | 1,5 | 218,0 | 231,5 | 322 | 825 | 80 | 1440 |
| NNC4844V | 6,4 | 220 | 270 | 50 | 1,5 | 239,0 | 252,5 | 338 | 906 | 85 | 1320 |
| NNC4848V | 10 | 240 | 300 | 60 | 2 | 259,5 | 276,5 | 506 | 1310 | 124 | 1200 |
| NNC4852V | 11 | 260 | 320 | 60 | 2 | 282,0 | 299,0 | 529 | 1430 | 132 | 1120 |
| NNC4856V | 16 | 280 | 350 | 69 | 2 | 307,0 | 326,5 | 691 | 1890 | 173 | 1030 |
| NNC4860V | 23 | 300 | 380 | 80 | 2,1 | 328,0 | 350,0 | 810 | 2170 | 196 | 950 |
| NNC4864V | 24 | 320 | 400 | 80 | 2,1 | 351,0 | 373,0 | 840 | 2330 | 207 | 900 |
| NNC4868V | 25,5 | 340 | 420 | 80 | 2,1 | 368,0 | 390,0 | 861 | 2450 | 215 | 850 |
| NNC4872V | 27 | 360 | 440 | 80 | 2,1 | 391,0 | 413,0 | 889 | 2600 | 225 | 810 |
| NNC4876V | 46 | 380 | 480 | 100 | 2,1 | 419,0 | 447,0 | 1300 | 3660 | 315 | 750 |
| NNC4880V | 48 | 400 | 500 | 100 | 2,1 | 433,5 | 461,5 | 1330 | 3790 | 325 | 720 |
| NNC4884V | 50 | 420 | 520 | 100 | 2,1 | 455,5 | 483,5 | 1360 | 3990 | 315 | 690 |
| NNC4888V | 52 | 440 | 540 | 100 | 2,1 | 477,5 | 506,0 | 1400 | 4180 | 320 | 660 |
| NNC4892V | 76 | 460 | 580 | 118 | 3 | 503,0 | 534,0 | 1570 | 4680 | 330 | 630 |
| NNC4896V | 79 | 480 | 600 | 118 | 3 | 527,0 | 558,0 | 1610 | 4900 | 350 | 600 |
| NNC48/500V | 82 | 500 | 620 | 118 | 3 | 543,0 | 574,0 | 1640 | 5060 | 380 | 580 |
| NNC48/530V | 86 | 530 | 650 | 118 | 3 | 575,0 | 606,0 | 1690 | 5360 | 390 | 550 |

**Series**
**NNC 49xx.V**

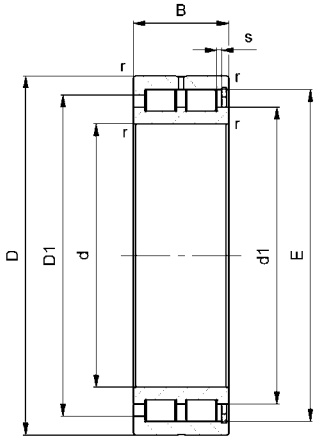





| | | measures [mm] | | | | | | load ratings | | fatigue limit load | limiting speed |
|---|---|---|---|---|---|---|---|---|---|---|---|
| description | weight [kg] | d | D | B | r min. | $d_1$ ca. | $D_1$ ca. | dyn. $C_r$ [kN] | stat. $C_{or}$ [kN] | $C_{ur}$ [kN] | $n_G$ [$min^{-1}$] |
| NNC4912V | 0,49 | 60 | 85 | 25 | 1 | 69,8 | 75,9 | 75 | 133 | 15,7 | 4450 |
| NNC4914V | 0,78 | 70 | 100 | 30 | 1 | 81,0 | 89,0 | 107 | 198 | 23,5 | 3800 |
| NNC4916V | 0,88 | 80 | 110 | 30 | 1 | 89,9 | 97,9 | 113 | 220 | 26 | 3400 |
| NNC4918V | 1,4 | 90 | 125 | 35 | 1,1 | 102,6 | 112,0 | 152 | 307 | 36 | 3000 |
| NNC4920V | 2 | 100 | 140 | 40 | 1,1 | 115,3 | 126,5 | 194 | 400 | 45 | 2700 |
| NNC4922V | 2,2 | 110 | 150 | 40 | 1,1 | 124,0 | 135,0 | 202 | 431 | 47 | 2490 |
| NNC4924V | 3 | 120 | 165 | 45 | 1,1 | 137,5 | 150,0 | 226 | 479 | 52 | 2270 |
| NNC4926V | 4 | 130 | 180 | 50 | 1,5 | 147,5 | 161,5 | 262 | 554 | 60 | 2090 |
| NNC4928V | 4,2 | 140 | 190 | 50 | 1,5 | 158,0 | 172,0 | 272 | 595 | 63 | 1960 |
| NNC4930V | 6,7 | 150 | 210 | 60 | 2 | 170,5 | 186,5 | 389 | 858 | 87 | 1800 |
| NNC4932V | 7 | 160 | 220 | 60 | 2 | 183,0 | 199,0 | 404 | 922 | 92 | 1710 |
| NNC4934V | 7,4 | 170 | 230 | 60 | 2 | 191,0 | 207,0 | 414 | 965 | 95 | 1620 |
| NNC4936V | 10,8 | 180 | 250 | 69 | 2 | 204,5 | 224,5 | 557 | 1240 | 126 | 1510 |
| NNC4938V | 11,2 | 190 | 260 | 69 | 2 | 215,0 | 234,5 | 572 | 1310 | 132 | 1440 |
| NNC4940V | 15,8 | 200 | 280 | 80 | 2,1 | 230,5 | 252,5 | 671 | 1510 | 153 | 1350 |
| NNC4944V | 17,2 | 220 | 300 | 80 | 2,1 | 247,5 | 269,5 | 697 | 1630 | 160 | 1250 |
| NNC4948V | 18,5 | 240 | 320 | 80 | 2,1 | 270,5 | 292,5 | 731 | 1780 | 171 | 1160 |
| NNC4952V | 32 | 260 | 360 | 100 | 2,1 | 294,5 | 322,5 | 1080 | 2540 | 243 | 1050 |
| NNC4956V | 34 | 280 | 380 | 100 | 2,1 | 316,5 | 344,5 | 1120 | 2740 | 255 | 980 |
| NNC4960V | 53 | 300 | 420 | 118 | 3 | 343,5 | 378,5 | 1560 | 3660 | 350 | 900 |
| NNC4964V | 56 | 320 | 440 | 118 | 3 | 362,0 | 397,0 | 1610 | 3860 | 365 | 850 |
| NNC4968V | 59 | 340 | 460 | 118 | 3 | 380,5 | 415,5 | 1650 | 4070 | 380 | 810 |
| NNC4972V | 62 | 360 | 480 | 118 | 3 | 399,0 | 434,0 | 1700 | 4270 | 395 | 770 |
| NNC4976V | 92,5 | 380 | 520 | 140 | 4 | 430,0 | 469,0 | 2210 | 5750 | 505 | 720 |
| NNC4980V | 96,5 | 400 | 540 | 140 | 4 | 450,5 | 489,0 | 2270 | 6030 | 525 | 690 |
| NNC4984V | 99,5 | 420 | 560 | 140 | 4 | 471,0 | 510,0 | 2330 | 6310 | 550 | 660 |
| NNC4988V | 137 | 440 | 600 | 160 | 4 | 499,5 | 548,0 | 2980 | 7540 | 565 | 630 |
| NNC4992V | 140 | 460 | 620 | 160 | 4 | 512,0 | 561,0 | 3020 | 7740 | 575 | 600 |
| NNC4996V | 165 | 480 | 650 | 170 | 5 | 537,0 | 589,0 | 3310 | 8560 | 710 | 570 |
| NNC49/500V | 175 | 500 | 670 | 170 | 5 | 564,0 | 615,0 | 3400 | 9000 | 745 | 550 |
| NNC49/530V | 200 | 530 | 710 | 180 | 5 | 589,0 | 645,0 | 3820 | 9910 | 810 | 520 |

**Series**
**NNCF 48xx.V**



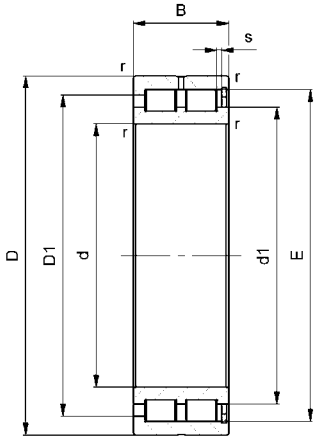



| | | measures [mm] | | | | | | | | load ratings | | fatigue limit load | limiting speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| description | weight [kg] | d | D | B | r min. | E | $d_1$ ca. | $D_1$ ca. | $s^{1)}$ | dyn. $C_r$ [kN] | stat. $C_{or}$ [kN] | $C_{ur}$ [kN] | $n_G$ [min$^{-1}$] |
| NNCF4830V | 2,9 | 150 | 190 | 40 | 1,1 | 178,5 | 164,0 | 175,0 | 2 | 234 | 575 | 58 | 1910 |
| NNCF4832V | 3,1 | 160 | 200 | 40 | 1,1 | 190,0 | 175,5 | 186,5 | 2 | 243 | 616 | 61 | 1800 |
| NNCF4834V | 4,1 | 170 | 215 | 45 | 1,1 | 201,6 | 185,5 | 197,5 | 3 | 265 | 651 | 64 | 1680 |
| NNCF4836V | 4,3 | 180 | 225 | 45 | 1,1 | 211,1 | 195,0 | 207,5 | 3 | 272 | 685 | 67 | 1600 |
| NNCF4838V | 5,7 | 190 | 240 | 50 | 1,5 | 225,2 | 207,5 | 221,0 | 4 | 314 | 784 | 77 | 1510 |
| NNCF4840V | 5,9 | 200 | 250 | 50 | 1,5 | 235,7 | 218,0 | 231,5 | 4 | 322 | 825 | 80 | 1440 |
| NNCF4844V | 6,4 | 220 | 270 | 50 | 1,5 | 256,8 | 239,0 | 252,5 | 4 | 338 | 906 | 85 | 1320 |
| NNCF4848V | 10 | 240 | 300 | 60 | 2 | 282,1 | 259,5 | 276,5 | 4 | 506 | 1310 | 124 | 1200 |
| NNCF4852V | 11 | 260 | 320 | 60 | 2 | 304,5 | 282,0 | 299,0 | 4 | 529 | 1430 | 132 | 1120 |
| NNCF4856V | 16 | 280 | 350 | 69 | 2 | 332,6 | 307,0 | 326,5 | 4 | 691 | 1890 | 173 | 1030 |
| NNCF4860V | 23 | 300 | 380 | 80 | 2,1 | 357,0 | 328,0 | 350,0 | 6 | 810 | 2170 | 196 | 950 |
| NNCF4864V | 24 | 320 | 400 | 80 | 2,1 | 379,9 | 351,0 | 373,0 | 6 | 840 | 2330 | 207 | 900 |
| NNCF4868V | 25,5 | 340 | 420 | 80 | 2,1 | 397,2 | 368,0 | 390,0 | 6 | 861 | 2450 | 215 | 850 |
| NNCF4872V | 27 | 360 | 440 | 80 | 2,1 | 420,1 | 391,0 | 413,0 | 6 | 889 | 2600 | 225 | 810 |
| NNCF4876V | 46 | 380 | 480 | 100 | 2,1 | 456,1 | 419,0 | 447,0 | 6 | 1300 | 3660 | 315 | 750 |
| NNCF4880V | 48 | 400 | 500 | 100 | 2,1 | 470,8 | 433,5 | 461,5 | 6 | 1330 | 3790 | 325 | 720 |
| NNCF4884V | 50 | 420 | 520 | 100 | 2,1 | 492,8 | 455,5 | 483,5 | 6 | 1360 | 3990 | 315 | 690 |
| NNCF4888V | 52 | 440 | 540 | 100 | 2,1 | 515,0 | 477,5 | 506,0 | 6 | 1400 | 4180 | 320 | 660 |
| NNCF4892V | 76 | 460 | 580 | 118 | 3 | 543,5 | 503,0 | 534,0 | 7 | 1570 | 4680 | 330 | 630 |
| NNCF4896V | 79 | 480 | 600 | 118 | 3 | 567,5 | 527,0 | 558,0 | 7 | 1610 | 4900 | 350 | 600 |
| NNCF48/500V | 82 | 500 | 620 | 118 | 3 | 583,5 | 543,0 | 574,0 | 7 | 1640 | 5060 | 380 | 580 |
| NNCF48/530V | 86 | 530 | 650 | 118 | 3 | 615,5 | 575,0 | 606,0 | 7 | 1690 | 5360 | 390 | 550 |

[1)] axial displacement facility from central position

**Series**
**NNCF 49xx.V**



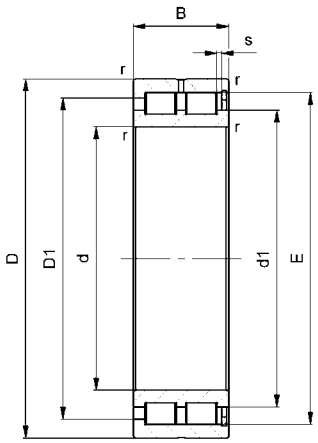



| | | measures [mm] | | | | | | | | load ratings | | fatigue limit load | limiting speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| description | weight [kg] | d | D | B | r min. | E | $d_1$ ca. | $D_1$ ca. | $s^{1)}$ | dyn. $C_r$ [kN] | stat. $C_{or}$ [kN] | $C_{ur}$ [kN] | $n_G$ [$min^{-1}$] |
| NNCF4912V | 0,49 | 60 | 85 | 25 | 1 | 78,9 | 69,8 | 75,9 | 1 | 75 | 133 | 15,7 | 4450 |
| NNCF4914V | 0,78 | 70 | 100 | 30 | 1 | 92,0 | 81,0 | 89,0 | 1 | 107 | 198 | 23,5 | 3800 |
| NNCF4916V | 0,88 | 80 | 110 | 30 | 1 | 100,9 | 89,9 | 97,9 | 1 | 113 | 220 | 26 | 3400 |
| NNCF4918V | 1,4 | 90 | 125 | 35 | 1,1 | 115,3 | 102,6 | 112,0 | 1,5 | 152 | 307 | 36 | 3000 |
| NNCF4920V | 2 | 100 | 140 | 40 | 1,1 | 129,7 | 115,3 | 126,5 | 2 | 194 | 400 | 45 | 2700 |
| NNCF4922V | 2,2 | 110 | 150 | 40 | 1,1 | 138,3 | 124,0 | 135,0 | 2 | 202 | 431 | 47 | 2490 |
| NNCF4924V | 3 | 120 | 165 | 45 | 1,1 | 153,7 | 137,5 | 150,0 | 3 | 226 | 479 | 52 | 2270 |
| NNCF4926V | 4 | 130 | 180 | 50 | 1,5 | 165,6 | 147,5 | 161,5 | 4 | 262 | 554 | 60 | 2090 |
| NNCF4928V | 4,2 | 140 | 190 | 50 | 1,5 | 176,1 | 158,0 | 172,0 | 4 | 272 | 595 | 63 | 1960 |
| NNCF4930V | 6,7 | 150 | 210 | 60 | 2 | 191,5 | 170,5 | 186,5 | 4 | 389 | 858 | 87 | 1800 |
| NNCF4932V | 7 | 160 | 220 | 60 | 2 | 203,9 | 183,0 | 199,0 | 4 | 404 | 922 | 92 | 1710 |
| NNCF4934V | 7,4 | 170 | 230 | 60 | 2 | 212,2 | 191,0 | 207,0 | 4 | 414 | 965 | 95 | 1620 |
| NNCF4936V | 10,8 | 180 | 250 | 69 | 2 | 230,6 | 204,5 | 224,5 | 4 | 557 | 1240 | 126 | 1510 |
| NNCF4938V | 11,2 | 190 | 260 | 69 | 2 | 240,8 | 215,0 | 234,5 | 4 | 572 | 1310 | 132 | 1440 |
| NNCF4940V | 15,8 | 200 | 280 | 80 | 2,1 | 259,4 | 230,5 | 252,5 | 5 | 671 | 1510 | 153 | 1350 |
| NNCF4944V | 17,2 | 220 | 300 | 80 | 2,1 | 276,6 | 247,5 | 269,5 | 5 | 697 | 1630 | 160 | 1250 |
| NNCF4948V | 18,5 | 240 | 320 | 80 | 2,1 | 299,6 | 270,5 | 292,5 | 5 | 731 | 1780 | 171 | 1160 |
| NNCF4952V | 32 | 260 | 360 | 100 | 2,1 | 331,5 | 294,5 | 322,5 | 6 | 1080 | 2540 | 243 | 1050 |
| NNCF4956V | 34 | 280 | 380 | 100 | 2,1 | 353,4 | 316,5 | 344,5 | 6 | 1120 | 2740 | 255 | 980 |
| NNCF4960V | 53 | 300 | 420 | 118 | 3 | 390,2 | 343,5 | 378,5 | 6 | 1560 | 3660 | 350 | 900 |
| NNCF4964V | 56 | 320 | 440 | 118 | 3 | 408,7 | 362,0 | 397,0 | 6 | 1610 | 3860 | 365 | 850 |
| NNCF4968V | 59 | 340 | 460 | 118 | 3 | 427,1 | 380,5 | 415,5 | 6 | 1650 | 4070 | 380 | 810 |
| NNCF4972V | 62 | 360 | 480 | 118 | 3 | 445,6 | 399,0 | 434,0 | 6 | 1700 | 4270 | 395 | 770 |
| NNCF4976V | 92,5 | 380 | 520 | 140 | 4 | 481,5 | 430,0 | 469,0 | 7 | 2210 | 5750 | 505 | 720 |
| NNCF4980V | 96,5 | 400 | 540 | 140 | 4 | 502,0 | 450,5 | 489,0 | 7 | 2270 | 6030 | 525 | 690 |
| NNCF4984V | 99,5 | 420 | 560 | 140 | 4 | 522,5 | 471,0 | 510,0 | 7 | 2330 | 6310 | 550 | 660 |
| NNCF4988V | 137 | 440 | 600 | 160 | 4 | 564,0 | 499,5 | 548,0 | 7 | 2980 | 7540 | 565 | 630 |
| NNCF4992V | 140 | 460 | 620 | 160 | 4 | 576,5 | 512,0 | 561,0 | 7 | 3020 | 7740 | 575 | 600 |
| NNCF4996V | 165 | 480 | 650 | 170 | 5 | 605,5 | 537,0 | 589,0 | 8 | 3310 | 8560 | 710 | 570 |
| NNCF49/500V | 175 | 500 | 670 | 170 | 5 | 632,0 | 564,0 | 615,0 | 8 | 3400 | 9000 | 745 | 550 |
| NNCF49/530V | 200 | 530 | 710 | 180 | 5 | 663,0 | 589,0 | 645,0 | 8 | 3820 | 9910 | 810 | 520 |

[1)] axial displacement facility from central position

## Series
## NNCF 50xx.V



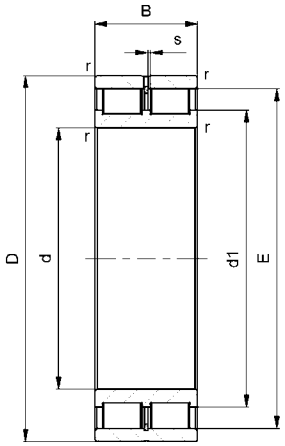



| | | measures [mm] | | | | | | | | load ratings | | fatigue limit load | limiting speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| description | weight [kg] | d | D | B | r min. | E | $d_1$ ca. | $D_1$ ca. | s[1)] | dyn. $C_r$ [kN] | stat. $C_{or}$ [kN] | $C_{ur}$ [kN] | $n_G$ [$min^{-1}$] |
| NNCF5004V | 0,2 | 20 | 42 | 30 | 0,6 | 37,9 | 27,5 | 35,3 | 4,5 | 49 | 56 | 6,8 | 10000 |
| NNCF5005V | 0,23 | 25 | 47 | 30 | 0,6 | 42,0 | 31,6 | 39,3 | 4,5 | 54 | 65 | 8,2 | 9000 |
| NNCF5006V | 0,35 | 30 | 55 | 34 | 1 | 49,7 | 38,5 | 46,8 | 4,5 | 69 | 90 | 11 | 7500 |
| NNCF5007V | 0,46 | 35 | 62 | 36 | 1 | 55,6 | 43,6 | 52,5 | 4,5 | 84 | 114 | 14 | 6700 |
| NNCF5008V | 0,56 | 40 | 68 | 38 | 1 | 61,8 | 49,0 | 58,5 | 4,5 | 98 | 138 | 17 | 6000 |
| NNCF5009V | 0,71 | 45 | 75 | 40 | 1 | 68,4 | 54,8 | 65,0 | 4,5 | 123 | 184 | 22,5 | 5300 |
| NNCF5010V | 0,76 | 50 | 80 | 40 | 1 | 73,8 | 60,2 | 70,4 | 4,5 | 130 | 203 | 23,6 | 5000 |
| NNCF5011V | 1,16 | 55 | 90 | 46 | 1,1 | 83,6 | 67,6 | 79,6 | 4,5 | 169 | 262 | 33,5 | 4300 |
| NNCF5012V | 1,24 | 60 | 95 | 46 | 1,1 | 86,8 | 70,8 | 82,8 | 4,5 | 174 | 276 | 35 | 4000 |
| NNCF5013V | 1,32 | 65 | 100 | 46 | 1,1 | 93,2 | 77,2 | 89,1 | 4,5 | 183 | 302 | 38,5 | 3800 |
| NNCF5014V | 1,85 | 70 | 110 | 54 | 1,1 | 100,4 | 81,1 | 95,5 | 5 | 226 | 357 | 46,5 | 3600 |
| NNCF5015V | 1,93 | 75 | 115 | 54 | 1,1 | 108,0 | 88,8 | 103,2 | 5 | 239 | 393 | 51 | 3200 |
| NNCF5016V | 2,59 | 80 | 125 | 60 | 1,1 | 117,1 | 94,6 | 111,4 | 5 | 289 | 460 | 57,5 | 3000 |
| NNCF5017V | 2,72 | 85 | 130 | 60 | 1,1 | 121,5 | 99,1 | 115,9 | 5 | 297 | 484 | 59,5 | 3000 |
| NNCF5018V | 3,62 | 90 | 140 | 67 | 1,5 | 130,2 | 106,2 | 124,5 | 5 | 346 | 573 | 69,5 | 2800 |
| NNCF5020V | 3,94 | 100 | 150 | 67 | 1,5 | 139,7 | 115,7 | 134,0 | 6 | 364 | 628 | 74 | 2600 |
| NNCF5022V | 6,32 | 110 | 170 | 80 | 2 | 156,2 | 127,0 | 149,0 | 6 | 479 | 814 | 95 | 2200 |
| NNCF5024V | 6,77 | 120 | 180 | 80 | 2 | 167,7 | 138,5 | 160,5 | 6 | 505 | 891 | 102 | 2000 |
| NNCF5026V | 10,2 | 130 | 200 | 95 | 2 | 184,0 | 148,5 | 175,5 | 7 | 717 | 1250 | 153 | 1900 |
| NNCF5028V | 11,1 | 140 | 210 | 95 | 2 | 197,9 | 162,5 | 189,5 | 7 | 756 | 1370 | 154 | 1800 |
| NNCF5030V | 13,3 | 150 | 225 | 100 | 2 | 206,9 | 170,0 | 198,0 | 7 | 793 | 1430 | 160 | 1700 |
| NNCF5032V | 16,2 | 160 | 240 | 109 | 2,1 | 224,9 | 184,5 | 215,0 | 7 | 909 | 1660 | 172 | 1500 |
| NNCF5034V | 23 | 170 | 260 | 122 | 2,1 | 243,0 | 198,0 | 232,0 | 7 | 1170 | 2160 | 224 | 1400 |
| NNCF5036V | 30,5 | 180 | 280 | 136 | 2,1 | 260,4 | 212,0 | 248,5 | 8 | 1350 | 2540 | 255 | 1300 |
| NNCF5038V | 31,5 | 190 | 290 | 136 | 2,1 | 269,9 | 221,5 | 258,0 | 8 | 1390 | 2660 | 265 | 1300 |
| NNCF5040V | 41 | 200 | 310 | 150 | 2,1 | 287,9 | 236,5 | 275,0 | 8 | 1590 | 3080 | 300 | 1200 |
| NNCF5044V | 52,5 | 220 | 340 | 160 | 3 | 312,4 | 254,5 | 298,0 | 8 | 1900 | 3680 | 355 | 1100 |
| NNCF5048V | 56 | 240 | 360 | 160 | 3 | 335,3 | 277,5 | 321,0 | 9,4 | 2000 | 4030 | 380 | 1000 |
| NNCF5052V | 85,5 | 260 | 400 | 190 | 4 | 376,2 | 304,0 | 358,5 | 9,4 | 2720 | 5270 | 475 | 900 |
| NNCF5056V | 90,5 | 280 | 420 | 190 | 4 | 390,5 | 318,0 | 372,5 | 9,4 | 2800 | 5540 | 500 | 850 |
| NNCF5060V | 130 | 300 | 460 | 218 | 4 | 431,8 | 354,5 | 412,5 | 9,4 | 3430 | 7110 | 630 | 750 |
| NNCF5064V | 135 | 320 | 480 | 218 | 4 | 447,1 | 370,0 | 428,0 | 9,4 | 3510 | 7440 | 590 | 700 |
| NNCF5068V | 185 | 340 | 520 | 243 | 5 | 481,6 | 401,5 | 461,5 | 9,4 | 4140 | 9140 | 670 | 670 |
| NNCF5072V | 195 | 360 | 540 | 243 | 5 | 503,0 | 416,0 | 481,5 | 9,4 | 4390 | 9460 | 725 | 630 |
| NNCF5076V | 200 | 380 | 560 | 243 | 5 | 520,0 | 433,5 | 498,5 | 9,4 | 4490 | 9870 | 730 | 600 |
| NNCF5080V | 270 | 400 | 600 | 272 | 5 | 559,0 | 462,5 | 535,0 | 9,4 | 5320 | 11640 | 870 | 560 |

[1)] axial displacement facility from central position

## Series
## NNCL 48xx.V



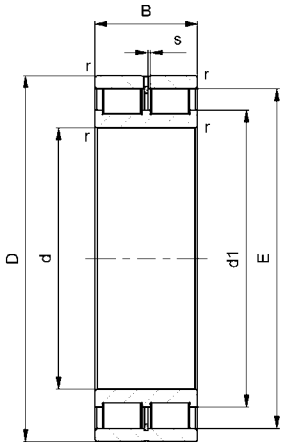



| | | measures [mm] | | | | | | | load ratings | | fatigue limit load | limiting speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| description | weight [kg] | d | D | B | r min. | E | $d_1$ ca. | s[1] | dyn. $C_r$ [kN] | stat. $C_{or}$ [kN] | $C_{ur}$ [kN] | $n_G$ [$min^{-1}$] |
| NNCL4830V | 2,9 | 150 | 190 | 40 | 1,1 | 178,5 | 164,0 | 2 | 234 | 575 | 58 | 1910 |
| NNCL4832V | 3,1 | 160 | 200 | 40 | 1,1 | 190,0 | 175,5 | 2 | 243 | 616 | 61 | 1800 |
| NNCL4834V | 4,1 | 170 | 215 | 45 | 1,1 | 201,6 | 185,5 | 3 | 265 | 651 | 64 | 1680 |
| NNCL4836V | 4,3 | 180 | 225 | 45 | 1,1 | 211,1 | 195,0 | 3 | 272 | 685 | 67 | 1600 |
| NNCL4838V | 5,7 | 190 | 240 | 50 | 1,5 | 225,2 | 207,5 | 4 | 314 | 784 | 77 | 1510 |
| NNCL4840V | 5,9 | 200 | 250 | 50 | 1,5 | 235,7 | 218,0 | 4 | 322 | 825 | 80 | 1440 |
| NNCL4844V | 6,4 | 220 | 270 | 50 | 1,5 | 256,8 | 239,0 | 4 | 338 | 906 | 85 | 1320 |
| NNCL4848V | 10 | 240 | 300 | 60 | 2 | 282,1 | 259,5 | 4 | 506 | 1310 | 124 | 1200 |
| NNCL4852V | 11 | 260 | 320 | 60 | 2 | 304,5 | 282,0 | 4 | 529 | 1430 | 132 | 1120 |
| NNCL4856V | 16 | 280 | 350 | 69 | 2 | 332,6 | 307,0 | 4 | 691 | 1890 | 173 | 1030 |
| NNCL4860V | 23 | 300 | 380 | 80 | 2,1 | 357,0 | 328,0 | 6 | 810 | 2170 | 196 | 950 |
| NNCL4864V | 24 | 320 | 400 | 80 | 2,1 | 379,9 | 351,0 | 6 | 840 | 2330 | 207 | 900 |
| NNCL4868V | 25,5 | 340 | 420 | 80 | 2,1 | 397,2 | 368,0 | 6 | 861 | 2450 | 215 | 850 |
| NNCL4872V | 27 | 360 | 440 | 80 | 2,1 | 420,1 | 391,0 | 6 | 889 | 2600 | 225 | 810 |
| NNCL4876V | 46 | 380 | 480 | 100 | 2,1 | 456,1 | 419,0 | 6 | 1300 | 3660 | 315 | 750 |
| NNCL4880V | 48 | 400 | 500 | 100 | 2,1 | 470,8 | 433,5 | 6 | 1330 | 3790 | 325 | 720 |
| NNCL4884V | 50 | 420 | 520 | 100 | 2,1 | 492,8 | 455,5 | 6 | 1360 | 3990 | 315 | 690 |
| NNCL4888V | 52 | 440 | 540 | 100 | 2,1 | 515,0 | 477,5 | 6 | 1400 | 4180 | 320 | 660 |
| NNCL4892V | 76 | 460 | 580 | 118 | 3 | 543,5 | 503,0 | 7 | 1570 | 4680 | 330 | 630 |
| NNCL4896V | 79 | 480 | 600 | 118 | 3 | 567,5 | 527,0 | 7 | 1610 | 4900 | 350 | 600 |
| NNCL48/500V | 82 | 500 | 620 | 118 | 3 | 583,5 | 543,0 | 7 | 1640 | 5060 | 380 | 580 |
| NNCL48/530V | 86 | 530 | 650 | 118 | 3 | 615,5 | 575,0 | 7 | 1690 | 5360 | 390 | 550 |

[1] axial displacement facility from central position

## Series
## NNCL 49xx.V



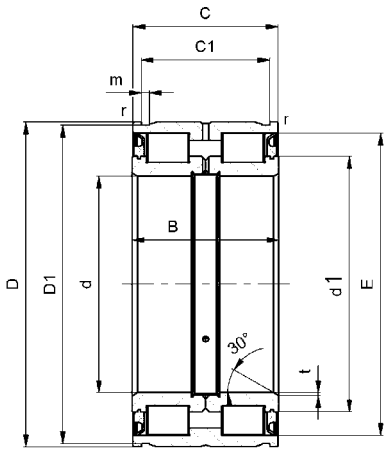



| | | measures [mm] | | | | | | | load ratings | | fatigue limit load | limiting speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| description | weight [kg] | d | D | B | r min. | E | $d_1$ ca. | s[1)] | dyn. $C_r$ [kN] | stat. $C_{or}$ [kN] | $C_{ur}$ [kN] | $n_G$ [$min^{-1}$] |
| NNCL4912V | 0,49 | 60 | 85 | 25 | 1 | 78,9 | 69,8 | 1 | 75 | 133 | 15,7 | 4450 |
| NNCL4914V | 0,78 | 70 | 100 | 30 | 1 | 92,0 | 81,0 | 1 | 107 | 198 | 23,5 | 3800 |
| NNCL4916V | 0,88 | 80 | 110 | 30 | 1 | 100,9 | 89,9 | 1 | 113 | 220 | 26 | 3400 |
| NNCL4918V | 1,4 | 90 | 125 | 35 | 1,1 | 115,3 | 102,6 | 1,5 | 152 | 307 | 36 | 3000 |
| NNCL4920V | 2 | 100 | 140 | 40 | 1,1 | 129,7 | 115,3 | 2 | 194 | 400 | 45 | 2700 |
| NNCL4922V | 2,2 | 110 | 150 | 40 | 1,1 | 138,3 | 124,0 | 2 | 202 | 431 | 47 | 2490 |
| NNCL4924V | 3 | 120 | 165 | 45 | 1,1 | 153,7 | 137,5 | 3 | 226 | 479 | 52 | 2270 |
| NNCL4926V | 4 | 130 | 180 | 50 | 1,5 | 165,6 | 147,5 | 4 | 262 | 554 | 60 | 2090 |
| NNCL4928V | 4,2 | 140 | 190 | 50 | 1,5 | 176,1 | 158,0 | 4 | 272 | 595 | 63 | 1960 |
| NNCL4930V | 6,7 | 150 | 210 | 60 | 2 | 191,5 | 170,5 | 4 | 389 | 858 | 87 | 1800 |
| NNCL4932V | 7 | 160 | 220 | 60 | 2 | 203,9 | 183,0 | 4 | 404 | 922 | 92 | 1710 |
| NNCL4934V | 7,4 | 170 | 230 | 60 | 2 | 212,2 | 191,0 | 4 | 414 | 965 | 95 | 1620 |
| NNCL4936V | 10,8 | 180 | 250 | 69 | 2 | 230,6 | 204,5 | 4 | 557 | 1240 | 126 | 1510 |
| NNCL4938V | 11,2 | 190 | 260 | 69 | 2 | 240,8 | 215,0 | 4 | 572 | 1310 | 132 | 1440 |
| NNCL4940V | 15,8 | 200 | 280 | 80 | 2,1 | 259,4 | 230,5 | 5 | 671 | 1510 | 153 | 1350 |
| NNCL4944V | 17,2 | 220 | 300 | 80 | 2,1 | 276,6 | 247,5 | 5 | 697 | 1630 | 160 | 1250 |
| NNCL4948V | 18,5 | 240 | 320 | 80 | 2,1 | 299,6 | 270,5 | 5 | 731 | 1780 | 171 | 1160 |
| NNCL4952V | 32 | 260 | 360 | 100 | 2,1 | 331,5 | 294,5 | 6 | 1080 | 2540 | 243 | 1050 |
| NNCL4956V | 34 | 280 | 380 | 100 | 2,1 | 353,4 | 316,5 | 6 | 1120 | 2740 | 255 | 980 |
| NNCL4960V | 53 | 300 | 420 | 118 | 3 | 390,2 | 343,5 | 6 | 1560 | 3660 | 350 | 900 |
| NNCL4964V | 56 | 320 | 440 | 118 | 3 | 408,7 | 362,0 | 6 | 1610 | 3860 | 365 | 850 |
| NNCL4968V | 59 | 340 | 460 | 118 | 3 | 427,1 | 380,5 | 6 | 1650 | 4070 | 380 | 810 |
| NNCL4972V | 62 | 360 | 480 | 118 | 3 | 445,6 | 399,0 | 6 | 1700 | 4270 | 395 | 770 |
| NNCL4976V | 92,5 | 380 | 520 | 140 | 4 | 481,5 | 430,0 | 7 | 2210 | 5750 | 505 | 720 |
| NNCL4980V | 96,5 | 400 | 540 | 140 | 4 | 502,0 | 450,5 | 7 | 2270 | 6030 | 525 | 690 |
| NNCL4984V | 99,5 | 420 | 560 | 140 | 4 | 522,5 | 471,0 | 7 | 2330 | 6310 | 550 | 660 |
| NNCL4988V | 137 | 440 | 600 | 160 | 4 | 564,0 | 499,5 | 7 | 2980 | 7540 | 565 | 630 |
| NNCL4992V | 140 | 460 | 620 | 160 | 4 | 576,5 | 512,0 | 7 | 3020 | 7740 | 575 | 600 |
| NNCL4996V | 165 | 480 | 650 | 170 | 5 | 605,5 | 537,0 | 8 | 3310 | 8560 | 710 | 570 |
| NNCL49/500V | 175 | 500 | 670 | 170 | 5 | 632,0 | 564,0 | 8 | 3400 | 9000 | 745 | 550 |
| NNCL49/530V | 200 | 530 | 710 | 180 | 5 | 663,0 | 589,0 | 8 | 3820 | 9910 | 810 | 520 |

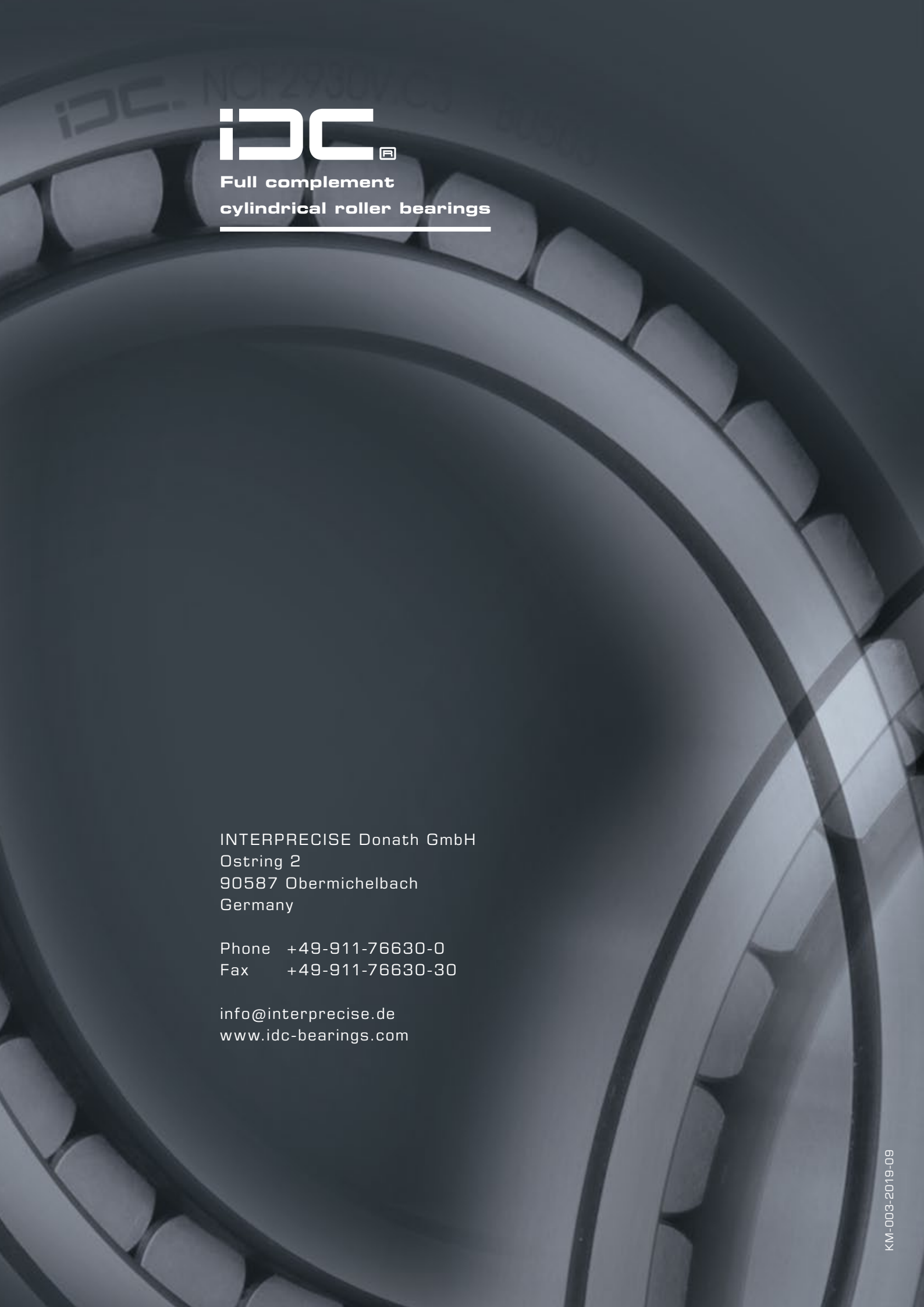
[1)] axial displacement facility from central position

## Series
## NNF 50xx.PP






| | | measures [mm] | | | | | | | | | | | load ratings | | fatigue limit load | limiting speed |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| description | weight [kg] | d | D | B | C | r min. | E | $d_1$ ca. | $C_1$ +0,2 | $D_1$ +0,2 | m | t | dyn. $C_r$ [kN] | stat. $C_{or}$ [kN] | $C_{ur}$ [kN] | $n_G$ [$min^{-1}$] |
| NNF5004.PP | 0,21 | 20 | 42 | 30 | 29 | 0,3 | 35,5 | 28,2 | 24,7 | 40,2 | 1,8 | 0,5 | 40 | 50 | 5,7 | 3600 |
| NNF5005.PP | 0,23 | 25 | 47 | 30 | 29 | 0,3 | 40,2 | 32,9 | 24,7 | 45,2 | 1,8 | 0,5 | 44 | 60 | 6,7 | 3000 |
| NNF5006.PP | 0,35 | 30 | 55 | 34 | 33 | 0,3 | 47,8 | 39,6 | 28,2 | 53 | 2,1 | 0,5 | 52 | 73 | 8,2 | 2600 |
| NNF5007.PP | 0,45 | 35 | 62 | 36 | 35 | 0,6 | 54,4 | 44,5 | 30,2 | 60 | 2,1 | 0,5 | 66 | 93 | 11,1 | 2200 |
| NNF5008.PP | 0,53 | 40 | 68 | 38 | 37 | 0,6 | 60,8 | 50,0 | 32,2 | 65,8 | 2,7 | 0,8 | 81 | 118 | 14,5 | 2000 |
| NNF5009.PP | 0,68 | 45 | 75 | 40 | 39 | 0,6 | 67,5 | 55,7 | 34,2 | 72,8 | 2,7 | 0,8 | 97 | 147 | 18,2 | 1800 |
| NNF5010.PP | 0,73 | 50 | 80 | 40 | 39 | 0,6 | 72,3 | 60,5 | 34,2 | 77,8 | 2,7 | 0,8 | 102 | 161 | 19,8 | 1700 |
| NNF5011.PP | 1,1 | 55 | 90 | 46 | 45 | 0,6 | 79,6 | 67,0 | 40,2 | 87,4 | 3,2 | 1,0 | 120 | 196 | 23,7 | 1500 |
| NNF5012.PP | 1,2 | 60 | 95 | 46 | 45 | 0,6 | 84,7 | 72,1 | 40,2 | 92,4 | 3,2 | 1,0 | 125 | 212 | 26 | 1400 |
| NNF5013.PP | 1,3 | 65 | 100 | 46 | 45 | 0,6 | 89,8 | 77,2 | 40,2 | 97,4 | 3,2 | 1,0 | 130 | 228 | 27,5 | 1300 |
| NNF5014.PP | 1,9 | 70 | 110 | 54 | 53 | 0,6 | 99,5 | 83,7 | 48,2 | 107,1 | 4,2 | 1,0 | 171 | 285 | 35 | 1200 |
| NNF5015.PP | 2 | 75 | 115 | 54 | 53 | 0,6 | 105,9 | 90,1 | 48,2 | 112,1 | 4,2 | 1,0 | 178 | 308 | 37,5 | 1100 |
| NNF5016.PP | 2,7 | 80 | 125 | 60 | 59 | 0,6 | 113,0 | 95,6 | 54,2 | 122,1 | 4,2 | 1,5 | 239 | 426 | 51 | 1000 |
| NNF5017.PP | 2,8 | 85 | 130 | 60 | 59 | 0,6 | 119,4 | 100,4 | 54,2 | 127,1 | 4,2 | 1,5 | 256 | 447 | 54 | 1000 |
| NNF5018.PP | 3,8 | 90 | 140 | 67 | 66 | 0,6 | 127,1 | 108,0 | 59,2 | 137 | 4,2 | 1,5 | 292 | 539 | 65 | 900 |
| NNF5019.PP | 4 | 95 | 145 | 67 | 66 | 0,6 | 130,9 | 111,9 | 59,2 | 142 | 4,2 | 1,5 | 297 | 559 | 66 | 900 |
| NNF5020.PP | 4,1 | 100 | 150 | 67 | 66 | 0,6 | 137,7 | 117,0 | 59,2 | 147 | 4,2 | 1,5 | 315 | 578 | 67 | 850 |
| NNF5022.PP | 6,5 | 110 | 170 | 80 | 79 | 0,6 | 154,1 | 130,0 | 70,2 | 167 | 4,2 | 1,8 | 389 | 711 | 82 | 750 |
| NNF5024.PP | 6,9 | 120 | 180 | 80 | 79 | 0,6 | 163,6 | 140,0 | 71,2 | 176 | 4,2 | 1,8 | 406 | 765 | 87 | 700 |
| NNF5026.PP | 10,5 | 130 | 200 | 95 | 94 | 0,6 | 183,0 | 153,0 | 83,2 | 196 | 4,2 | 1,8 | 582 | 1062 | 121 | 630 |
| NNF5028.PP | 11 | 140 | 210 | 95 | 94 | 0,6 | 195,1 | 165,0 | 83,2 | 206 | 5,2 | 1,8 | 608 | 1150 | 129 | 600 |
| NNF5030.PP | 13,5 | 150 | 225 | 100 | 99 | 0,6 | 209,0 | 175,5 | 87,2 | 221 | 5,2 | 2,0 | 697 | 1300 | 143 | 560 |
| NNF5032.PP | 16,5 | 160 | 240 | 109 | 108 | 0,6 | 222,4 | 189,0 | 95,2 | 236 | 5,2 | 2,0 | 727 | 1410 | 152 | 500 |
| NNF5034.PP | 22,5 | 170 | 260 | 122 | 121 | 0,6 | 238,8 | 200,5 | 107,2 | 254 | 5,2 | 2,0 | 934 | 1800 | 191 | 480 |
| NNF5036.PP | 30 | 180 | 280 | 136 | 135 | 0,6 | 258,7 | 217,5 | 118,2 | 274 | 5,2 | 2,0 | 1110 | 2180 | 231 | 450 |
| NNF5038.PP | 31,5 | 190 | 290 | 136 | 135 | 0,6 | 267,0 | 225,5 | 118,2 | 284 | 5,2 | 2,0 | 1130 | 2270 | 240 | 430 |
| NNF5040.PP | 42 | 200 | 310 | 150 | 149 | 0,6 | 283,6 | 242,0 | 128,2 | 304 | 6,3 | 2,0 | 1290 | 2740 | 275 | 400 |
| NNF5044.PP | 53,5 | 220 | 340 | 160 | 159 | 1 | 308,1 | 260,0 | 138,2 | 334 | 6,3 | 2,0 | 1520 | 3140 | 315 | 360 |
| NNF5048.PP | 57,5 | 240 | 360 | 160 | 159 | 1 | 327,2 | 279,5 | 138,2 | 354 | 6,3 | 2,0 | 1580 | 3380 | 335 | 340 |
| NNF5052.PP | 84,5 | 260 | 400 | 190 | 189 | 1,1 | 369,7 | 312,0 | 162,2 | 394 | 6,3 | 3,0 | 2200 | 4720 | 480 | 320 |
| NNF5056.PP | 90 | 280 | 420 | 190 | 189 | 1,1 | 392,6 | 335,0 | 163,2 | 413 | 7,3 | 3,0 | 2290 | 5080 | 510 | 300 |
| NNF5060.PP | 126 | 300 | 460 | 218 | 216 | 1,1 | 417,9 | 351,0 | 185,2 | 453 | 7,3 | 3,0 | 2880 | 6210 | 595 | 280 |
| light series: | | | | | | | | | | | | | | | | |
| NNF130.PP | 7,5 | 130 | 190 | 80 | 79 | 0,6 | 173,2 | 149,5 | 71,2 | 186 | 4,2 | 1,8 | 421 | 820 | 92 | 670 |
| NNF140.PP | 8 | 140 | 200 | 80 | 79 | 0,6 | 182,8 | 159,0 | 71,2 | 196 | 4,2 | 1,8 | 436 | 875 | 96 | 630 |
| NNF150.PP | 8,4 | 150 | 210 | 80 | 79 | 0,6 | 197,1 | 173,5 | 71,2 | 206 | 5,2 | 1,8 | 457 | 957 | 101 | 560 |
| NNF160.PP | 8,8 | 160 | 220 | 80 | 79 | 0,6 | 206,6 | 183,0 | 71,2 | 216 | 5,2 | 1,8 | 471 | 1010 | 106 | 530 |
| NNF170.PP | 9,3 | 170 | 230 | 80 | 79 | 0,6 | 216,2 | 192,5 | 71,2 | 226 | 5,2 | 1,8 | 484 | 1060 | 111 | 530 |
| NNF180.PP | 9,8 | 180 | 240 | 80 | 79 | 0,6 | 225,8 | 202,0 | 71,2 | 236 | 5,2 | 1,8 | 497 | 1120 | 115 | 500 |
| NNF190.PP | 12,7 | 190 | 260 | 80 | 79 | 0,6 | 240,1 | 216,5 | 73,2 | 254 | 5,2 | 1,8 | 516 | 1200 | 121 | 450 |
| NNF200.PP | 13,2 | 200 | 270 | 80 | 79 | 0,6 | 249,7 | 226,0 | 73,2 | 264 | 5,2 | 1,8 | 528 | 1260 | 125 | 430 |
| NNF220.PP | 19,5 | 220 | 300 | 95 | 94 | 1 | 276,6 | 248,0 | 83,2 | 294 | 5,2 | 2,0 | 698 | 1630 | 162 | 400 |
| NNF240.PP | 21 | 240 | 320 | 95 | 94 | 1 | 299,6 | 271,0 | 83,2 | 314 | 6,3 | 2,0 | 732 | 1780 | 170 | 370 |
| NNF260.PP | 22,5 | 260 | 340 | 95 | 94 | 1 | 322,3 | 292,0 | 83,2 | 334 | 6,3 | 3,0 | 833 | 1080 | 195 | 350 |
| NNF300.PP | 25,5 | 300 | 380 | 95 | 94 | 1 | 358,6 | 328,5 | 83,2 | 374 | 6,3 | 3,0 | 887 | 2350 | 205 | 320 |



# Full complement cylindrical roller bearings

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